

Efficient Pairing-Based Batch Arguments for NP with a Constant-Size Proof

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Abstract. Non-interactive Batch arguments (BARGs) for NP relations enable a prover to generate a single succinct proof for multiple NP instances, significantly amortizing verification costs. While recent pairing-based BARGs achieve impressive results, their practical efficiency remains limited by proof sizes and verifier pairing operations that scale linearly with the size of the Boolean circuit computing the NP relation. In this work, we present a novel pairing-based BARG construction that achieves constant proof size and a constant number of pairing operations, independent of both the number of instances and the circuit size. Our approach leverages a bivariate polynomial commitment to compactly encode all wire values across instances, and introduces a new efficient bi-to-uni variate sumcheck protocol, called BuLosum. BuLosum improves upon prior techniques by reducing proving costs to a single multi-scalar multiplication and verification to only three pairings. Using BuLosum, we further design an optimized matrix multiplication protocol that minimizes both proof size and verification overhead. By integrating these components, we obtain the first pairing-based BARG achieving constant-size proofs and constant number of pairing operations.

Keywords: Polynomial Commitment; Sumcheck Scheme; Matrix Multiplication; Batch Arguments

1 Introduction

Non-interactive batch arguments (BARGs) for an NP relation $\mathcal{R}$ [1,2,3,4] are protocols that enable an efficient prover to generate a single, succinct batch proof π that attests to the validity of ℓ instances $(\mathbf{x}_1, \dots, \mathbf{x}_\ell)$, given their corresponding witnesses $(\mathbf{w}_1, \dots, \mathbf{w}_\ell)$. This proof π is non-interactive and publicly verifiable, with proof size bounded by $\text{poly}(\lambda, |\mathcal{R}|, \log \ell)$, where λ is the security parameter and $|\mathcal{R}|$ is the size of the Boolean circuit computing the NP-relation $\mathcal{R}$. Soundness requires that if any instance is false, i.e., $\mathcal{R}(\mathbf{x}_i, \mathbf{w}_i) \neq 1$ for some $i \in [\ell]$, then no probabilistic polynomial-time (PPT) adversary can produce an accepting proof with non-negligible probability. Batch arguments not only amortize the cost of NP verification across many instances but also serve as a critical component in constructing various cryptographic primitives, such as RAM delegation [5], incrementally verifiable computation (IVC) [6,7,8], monotone policy aggregated signatures derived from monotone policy BARGs [9,10,11], multi-key homomorphic signatures [12,13], and non-interactive zero-knowledge proofs [14,15,16].

More recently, batch arguments have emerged as a unifying abstraction for constructing expressive and general-purpose succinct non-interactive arguments of knowledge (SNARGs) for broad classes of computations [17,18,19,20,21]. Notably, Jin et al.[21] proposed a universal SNARG construction via the encrypt-hash-and-BARG paradigm [22]. [18,19] introduced a general “BARG-to-SNARG compiler” showing that any BARG can be transformed into a SNARG for P and NTISP. In combination with BARG constructions based on the Learning With Errors (LWE) assumption [19], this yields SNARGs for P and NTISP (a complexity class of problems decidable by a nondeterministic Turing machine within simultaneous time and space bounds) from standard post-quantum assumptions. Furthermore, when instantiated with BARGs constructed from bilinear maps [1,5,23] or from the combination of the Decisional Diffie–Hellman (DDH) and Quadratic Residuosity (QR) assumptions [20], the compiler yields corresponding SNARGs under these classical assumptions.

Limitation in prover and verifier efficiency. Despite their expressiveness and utility, the practical efficiency of existing BARGs remains a major bottleneck—particularly in the common reference string (CRS) model, where a structured reference string is generated once during a trusted setup [24,23].

We focus on pairing-based batch arguments instantiated in this CRS model. A common limitation of existing constructions in this setting is that both the proof size and the number of verifier pairing operations scale linearly with the Boolean circuit size $|\mathcal{R}|$ that defines the underlying NP relation. Concretely, when verifying ℓ instances of an NP relation computable by a circuit of size $|\mathcal{R}|$, both the proof size and the number of pairing operations typically grow as $\text{poly}(\lambda, |\mathcal{R}|, \log \ell)$. Furthermore, in most known constructions, the CRS size and the prover complexity grow quadratically with the number of instances ℓ , except for the recent work of [24], which achieves a strictly linear-size CRS and quasi-linear prover complexity. Nevertheless, even in [24], both the number of pairing checks and the proof size remain linear in the circuit size $|\mathcal{R}|$, limiting scalability in large-scale batch verification settings.

A natural question arises: can the communication and verification overhead of batch arguments be reduced even further? We investigate the most extreme point on the efficiency spectrum by asking:

Can we construct a BARG whose proof size and verifier pairing operations are constant—independent of the number of instances ℓ and the circuit size $|\mathcal{R}|$?

Results. In this work, we put forward an efficient pairing-based batch argument for NP within the framework of [23], which achieves constant proof size and requires only a constant number of pairing operations.

We evaluate the theoretical performance of our proposed protocol and compare it with representative existing constructions, as summarized in Fig. 1. Both [24,25] and our scheme build upon the general framework of [23], but they make different efficiency trade-offs. Specifically, [24] improves upon [23] by reducing prover time and linear CRS. [25] shifts focus to witness size $|w|$ for proof size and pairing, which may be beneficial when $|w| \ll |C|$, but it suffers from significantly higher prover time and CRS size.

In contrast, our construction achieves a constant proof size and requires only a constant number of pairings — a unique improvement that establishes a new efficiency frontier for batch arguments. Our prover time, $O(\lambda \cdot \ell \cdot |C|)$, matches the best prior work [24]. Although we retain a quadratic CRS size of $O(\lambda \cdot \ell \cdot |C|)$, this is a favorable trade-off given the substantial gains in proof compactness and verification efficiency, making our scheme particularly suitable for applications where small proof size and low verification overhead are critical.

Scheme	Proof Size	Prover Time	Pairing	Linear CRS
[23]	$O(\lambda \cdot C)$	$O(\lambda \cdot \ell^2 \cdot C)$	$ C \cdot P$	$O(\lambda \cdot \ell^2)$
[24]	$O(\lambda \cdot C)$	$O(\lambda \cdot \ell \cdot C)$	$ C \cdot P$	$O(\lambda \cdot \ell)$
[25]	$O(\lambda \cdot w)$	$O(\lambda \cdot \ell^5 \cdot C ^5)$	$ w \cdot P$	$O(\lambda \cdot \ell^5 \cdot C ^5)$
Ours	$O(1)$	$O(\lambda \cdot \ell \cdot C)$	$O(1)P$	$O(\lambda \cdot \ell \cdot C)$

Fig. 1: Comparison of batch argument systems for NP. $|C|$ denotes the size of the circuit C , $|w|$ denotes the size of the witness w , P denotes pairing computations.

Strategy. As a core component of our construction, we introduce BuLosum, a new KZG-based bi-to-uni variate sumcheck that attains optimal proving cost and smaller proof size compared to the traditional bi-to-uni variate sumcheck [26]. Our main strategies are as follows.

1. We extend the efficient univariate sumcheck protocol Losum [27] to a new bi-to-uni variate sumcheck scheme, denoted BuLosum, which significantly improves upon the existing bi-to-uni variate sumcheck construction of [26] in the bivariate KZG setting. The prover cost consists of a single multi-scalar multiplication (MSM) of size k over $\mathbb{G}_1$, where k is the number of non-zero entries in the input matrix. The verification cost mainly requires three pairings, and the proof consists of a single $\mathbb{G}_1$ element.
2. We first design a matrix multiplication protocol, MatrixMul, using the traditional bi-to-uni variate sumcheck technique. Then, we employ BuLosum as a core building block to construct an improved protocol, MatrixMul+. The new protocol achieves significantly better ef-

efficiency: it reduces the proof size from $6|\mathbb{G}_1| + 2|\mathbb{G}_2|$ in MatrixMul to only $3|\mathbb{G}_1|$, and lowers the verification cost from 18 pairings to just 9 pairings.

3. Finally, we employ BuLosum and the resulting matrix multiplication protocol MatrixMul+ to develop a pairing-based batch argument for NP. This construction achieves both a constant-size proof and a constant number of pairing operations, leading to a significant reduction in verification cost.

Concurrent Work. A concurrent work [25] also achieves a compact proof size and verification time independent of the circuit size. Their techniques differ from ours. Their construction builds a BARG generically from a projective chainable functional commitment that satisfies somewhere extraction property, with proof size and online verifier runtime scaling as $O(\lambda|w|)$. However, our scheme is based on a bivariate polynomial commitment combined with a bi-to-univariate sumcheck protocol BuLosum. This allows us to achieve constant proof size and constant pairing-based verification time. While our result currently guarantees non-adaptive computational soundness, it provides significant efficiency advantages in terms of proof compactness and verification complexity.

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2 Technical Overview

We start by briefly reviewing the Waters–Wu batch argument [23], which serves as the conceptual starting point of our construction. This scheme is built over (symmetric) composite-order bilinear groups and realizes a non-interactive batch argument for Boolean circuit satisfiability. Let $\mathbb{G}$ be a group of order $N = pq$ equipped with a bilinear map $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$. The construction exploits the subgroup structure $\mathbb{G}_p, \mathbb{G}_q \subseteq \mathbb{G}$ of orders p and q , respectively, to separate soundness and hiding components.

Waters-Wu Construction [23]. For ℓ instances $(\mathbf{x}_1, \dots, \mathbf{x}_\ell)$ and associated witnesses $(\mathbf{w}_1, \dots, \mathbf{w}_\ell)$ of a Boolean circuit $C : \{0, 1\}^n \times \{0, 1\}^h \rightarrow \{0, 1\}$, the prover first commits to all wire values across instances. That is, for each wire $j \in [t]$, the prover forms a vector commitment

$$U_j = \prod_{i \in [\ell]} A_i^{v_{i,j}} = g_p^{\sum_{i \in [\ell]} \alpha_i v_{i,j}},$$

where $\{A_i = g_p^{\alpha_i}\}_{i \in [\ell]}$ are CRS elements and $v_{i,j} \in \{0, 1\}$ is the j -th wire value of $C(\mathbf{x}_i, \mathbf{w}_i)$. The proof then mainly consists of three types of validity checks:

- **Wire validity checks.** The prover constructs proof elements $V_j = \prod_{i \in [\ell]} \prod_{i' \neq i} B_{i,i'}^{w_{i,j} w_{i',j} - w_{i',j}}$ for $j \in [t]$ using cross terms $B_{i,i'} = g_p^{\alpha_i \alpha_{i'}}$ to convince the verifier that each U_j encodes Boolean values vector, i.e., $v_{i,j}^2 = v_{i,j}$. The verifier checks

$$e(U_j, U_j) = e(g_p, V_j) \cdot e\left(\prod_{i \in [\ell]} A_i, U_j\right).$$

- **Gate validity checks.** For each NAND gate $G_k = (k_1, k_2, k_3) \in [t]^3$ with $k \in [s]$, where k_1, k_2 denote the input wire indices and k_3 denotes the output wire index, the prover constructs proof elements W_k using cross terms to convince the verifier that $v_{i,k_3} = \text{NAND}(v_{i,k_1}, v_{i,k_2})$, i.e., $v_{i,k_3} = 1 - v_{i,k_1} v_{i,k_2}$. The verifier checks

$$\frac{e(\prod_{i \in [\ell]} A_i, U_{k_3}) e(U_{k_1}, U_{k_2})}{e(\prod_{i \in [\ell]} A_i, \prod_{i \in [\ell]} A_i)} = e(g_p, W_k).$$

- **Output consistency checks.** The verifier checks that the output commitment U_t is a commitment to the all-ones vector, i.e., $U_t = \prod_{i \in [\ell]} A_i$.

Limitation. The verifier must check pairing equations for all t wires and s gates, resulting in $O(|C|)$ proof size and $O(|C|)$ pairing operations. This linear overhead becomes a bottleneck in large-scale settings. Motivated by this limitation, we seek to design a scalable aggregation mechanism that reduces both the proof size and the number of pairing operations to constant, independent of $|C|$.

Our approach. The key insight of our work is to replace the explicit vector commitments in [23] with a bivariate polynomial commitment that jointly encodes all wire values across all instances. Instead of committing to each wire value vector separately using Pedersen-style vector commitment, we interpolate all wire values into a bivariate polynomial $\phi(X, Y)$ of degrees $\deg(X) < \ell, \deg(Y) < t$ that encodes the entire wire matrix over $\mathbb{L} \times \mathbb{V}$:

$$\mathbf{V}[i, j] = v_{i,j} = \phi(\omega_L^i, \omega_V^j),$$

where $\mathbb{L} = \{\omega_L^1, \dots, \omega_L^\ell\}$ and $\mathbb{V} = \{\omega_V^1, \dots, \omega_V^t\}$ correspond to instance indices and wire indices, respectively. The prover then commits to $\phi(X, Y)$ as $\phi = [\phi(\tau_X, \tau_Y)]$ using a bivariate KZG polynomial commitment [28, 29], where τ_X, τ_Y are trapdoors in the KZG CRS $\{[\tau_X^{i-1} \tau_Y^{j-1}]\}_{i \in [\ell], j \in [t]}$. Similarly, let $f^{(1)}(X, Z), f^{(2)}(X, Z), f^{(3)}(X, Z)$ be the unique bivariate polynomials of degrees $\deg_X < \ell$ and $\deg_Z < s$ that interpolate the gate matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ over $\mathbb{L} \times \mathbb{H}$, respectively. Here, $\mathbb{H}$ corresponds to the gate indices. The gate matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ correspond to the input and output wire values of all gates, where the k -th column of each matrix encodes the vectors $\mathbf{a}_k, \mathbf{b}_k, \mathbf{c}_k$ associated with k -th NAND gate $G_k = (\mathbf{a}_k, \mathbf{b}_k, \mathbf{c}_k)$ for $k \in [s]$. Formally,

$$\begin{aligned} \mathbf{A}[i, k] &= \mathbf{a}_k[i] = f^{(1)}(\omega_L^i, \omega_H^k), & \mathbf{B}[i, k] &= \mathbf{b}_k[i] = f^{(2)}(\omega_L^i, \omega_H^k), \\ \mathbf{C}[i, k] &= \mathbf{c}_k[i] = f^{(3)}(\omega_L^i, \omega_H^k). \end{aligned}$$

Here, $\mathbf{a}_k, \mathbf{b}_k$ denote the input wire value vectors of gate G_k and $\mathbf{c}_k$ denotes its output wire value vector.

This representation allows the prover to check both wire validity and gate validity through bivariate zero-test on the derived polynomials of $\phi(X, Y)$ and $\{f^{(i)}(X, Z)\}_{i \in [3]}$.

- **Wire validity check.** Verifying that $v_{i,j}^2 = v_{i,j}$ holds is reduced to verifying that all values of matrix $\mathbf{V}$ are Boolean, i.e., $\mathbf{V}^2[i, j] = \mathbf{V}[i, j]$ encoded as the polynomial identity

$$\Phi(X, Y) = \phi(X, Y)\phi(X, Y) - \phi(X, Y) = 0.$$

This is equivalent to proving that $\Phi(\omega_L^i, \omega_V^j) = \phi(\omega_L^i, \omega_V^j)\phi(\omega_L^i, \omega_V^j) - \phi(\omega_L^i, \omega_V^j) = 0$ for all $\omega_L^i \in \mathbb{L}, \omega_V^j \in \mathbb{V}$. The prover can prove this via a bivariate zero-test [26]. Let $\rho(R, Y)$ be a prescribed auxiliary polynomial such that all $\rho(R, \omega_V^j)$ for $\omega_V^j \in \mathbb{V}$ are $|\mathbb{V}|$ linearly independent polynomials. Let $\Phi^*(X, R) = \sum_{\omega_V^j \in \mathbb{V}} \Phi(X, \omega_V^j) \rho(R, \omega_V^j)$. The zero-test relation is enforced in the following two steps:

1. $\forall \omega_L^i \in \mathbb{L}, \Phi^*(\omega_L^i, R) = 0$. This relies on the following polynomial identity

$$\Phi^*(X, R) = \eta(X) \cdot \zeta_{\mathbb{L}}(X),$$

where $\zeta_{\mathbb{L}}(X)$ is a vanishing polynomial on the subgroup $\mathbb{L}$.

2. $\Phi^*(X, R) = \sum_{\omega_V^j \in \mathbb{V}} \Phi(X, \omega_V^j) \cdot \rho(R, \omega_V^j)$. This relies on the following polynomial identity (weighted bi-to-uni variate sumcheck lemma 1)

$$\Phi(X, Y) \cdot \rho(\tau_R, Y) = \frac{\Phi^*(X, \tau_R)}{|\mathbb{V}|} + Q(X, Y) \cdot Y + D(X, Y) \cdot \zeta_{\mathbb{V}}(Y),$$

where $Q(X, Y)$ is a polynomial of $\deg(X) < |\mathbb{L}|, \deg(Y) < |\mathbb{V}| - 1$ and $\zeta_{\mathbb{V}}(Y)$ is a vanishing polynomial on the subgroup $\mathbb{V}$.

The proof consists of the polynomial commitments of $Q(X, Y), D(X, Y)$ and $\eta(X)$, i.e. $\pi_1 = ([Q(\tau_X, \tau_Y)], [D(\tau_X, \tau_Y)], [\eta(\tau_X)])$.

- **Gate validity check.** Verifying that $v_{i,k_3} = 1 - v_{i,k_1}v_{i,k_2}$ holds is reduced to verifying that $\mathbf{C}[i, k] = \text{NAND}(\mathbf{A}[i, k], \mathbf{B}[i, k])$, i.e., $\mathbf{A}[i, k] \cdot \mathbf{B}[i, k] = 1 - \mathbf{C}[i, k]$ encoded as the following polynomial identity

$$G(X, Z) = 1 - f^{(1)}(X, Z)f^{(2)}(X, Z) - f^{(3)}(X, Z) = 0,$$

which is equivalent to proving that $G(\omega_L^i, \omega_H^k) = 1 - f^{(1)}(\omega_L^i, \omega_H^k)f^{(2)}(\omega_L^i, \omega_H^k) - f^{(3)}(\omega_L^i, \omega_H^k) = 0$ for all $\omega_L^i \in \mathbb{L}$ and $\omega_H^k \in \mathbb{H}$. Introducing $\bar{\rho}(R, Z)$ be a prescribed auxiliary polynomial such that all $\bar{\rho}(R, \omega_H^k)$ for $\omega_H^k \in \mathbb{H}$ are $|\mathbb{H}|$ linearly independent polynomials. Let $G^*(X, R) = \sum_{\omega_H^k \in \mathbb{H}} G(X, \omega_H^k) \bar{\rho}(R, \omega_H^k)$. The zero-test relation is enforced in the following two steps:

1. $\forall \omega_L^i \in \mathbb{L}, G^*(\omega_L^i, R) = 0$. This relies on the following identity

$$G^*(X, R) = \bar{\eta}(X) \cdot \zeta_{\mathbb{L}}(X).$$

2. $G^*(X, R) = \sum_{\omega_H^k \in \mathbb{H}} G(X, \omega_H^k) \bar{\rho}(R, \omega_H^k)$. This relies on the following identity

$$G(X, Z) \cdot \bar{\rho}(\tau_R, Z) = \frac{G^*(X, \tau_R)}{|\mathbb{H}|} + \bar{Q}(X, Z) \cdot Z + \bar{D}(X, Z) \cdot \zeta_{\mathbb{H}}(Z),$$

where $\bar{Q}(X, Z)$ is a polynomial of $\deg(X) < |\mathbb{L}|$, $\deg(Z) < |\mathbb{H}| - 1$ and $\zeta_{\mathbb{H}}(Z)$ is a vanishing polynomial on the subgroup $\mathbb{H}$.

The proof consists of the polynomial commitments of $\bar{Q}(X, Z)$, $\bar{D}(X, Z)$ and $\bar{\eta}(X)$, i.e., $\pi_2 = ([\bar{Q}(\tau_X, \tau_Z)], [\bar{D}(\tau_X, \tau_Z)], [\bar{\eta}(\tau_X)])$.

- **Output consistency checks.** Verifying that $U_t = \prod_{i \in [\ell]} A_i$ holds is reduced to checking that $\phi(X, t)$ is identically 1. Formally, this verification is performed by proving the opening relation check that

$$\phi(X, Y) - \phi_t(X) = Q_t(X, Y) \cdot (Y - t),$$

where $\phi_t(X) = \phi(X, t) \equiv 1$ and $Q_t(X, Y)$ is the quotient polynomial certifying divisibility by $(Y - t)$. The proof consists of the polynomial commitment of $Q_t(X, Y)$, i.e., $\pi_3 = [Q_t(\tau_X, \tau_Y)]$.

This unified representation allows us to perform aggregation and verification operations directly in the polynomial domain, achieving constant-size proofs and a constant number of pairing operations.

Challenge: Gate–Wire Consistency. While the bivariate encoding achieves compactness, it introduces a subtle challenge in maintaining consistency between gate and wire validity checks. These two types of constraints are tightly coupled: verifying gate correctness necessarily depends on the Booleanity of the corresponding wire values. However, when each set of values is encoded by separate bivariate polynomials—namely, the global wire matrix $\phi(X, Y)$ and the gate matrices $\{f^{(i)}(X, Z)\}_{i \in [3]}$ —this linkage is lost.

Gate permutation validity. To re-establish this consistency, we introduce an additional gate permutation check that binds the gate matrices **A**, **B**, **C** to the global wire matrix **V**. Concretely, we ensure that the gate input and output wire values used in the gate validity check are precisely those committed in the wire matrix.

This is achieved by enforcing that there exist selection matrices $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3 \in \{0, 1\}^{t \times s}$. These selection matrices $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$ are sparse matrices defined by the circuit wiring pattern. For each NAND gate $G_k = (\mathbf{a}_k, \mathbf{b}_k, \mathbf{c}_k)$, where $k \in [s]$, we define three index sets $I_1, I_2, I_3 \subseteq [t]$ that specify which columns of the global wire matrix $\mathbf{V} = (\mathbf{v}_1, \dots, \mathbf{v}_t)$ correspond to the gate’s left input, right input, and output vectors, respectively. The selection matrices are constructed as follows:

- $\mathbf{P}_1[\mu_1(k), k] = 1$ and all other entries are 0, where $\mu_1 : [s] \rightarrow I_1$ is an injective mapping such that $\mathbf{a}_k = \mathbf{v}_{\mu_1(k)}$ for all k . Similarly, $I_1 = \{\mu_1(k) \mid k \in [s]\}$.

- $\mathbf{P}_2[\mu_2(k), k] = 1$ and all other entries are 0, where $\mu_2 : [s] \rightarrow I_2$ is an injective mapping such that $\mathbf{b}_k = \mathbf{v}_{\mu_2(k)}$ for all k . Similarly, $I_2 = \{\mu_2(k) \mid k \in [s]\}$.
- $\mathbf{P}_3[\mu_3(k), k] = 1$ and all other entries are 0, where $\mu_3 : [s] \rightarrow I_3$ is an injective mapping such that $\mathbf{c}_k = \mathbf{v}_{\mu_3(k)}$ for all k . Similarly, $I_3 = \{\mu_3(k) \mid k \in [s]\}$.

Thus, we have:

$$\mathbf{A} = \mathbf{V}\mathbf{P}_1, \quad \mathbf{B} = \mathbf{V}\mathbf{P}_2, \quad \mathbf{C} = \mathbf{V}\mathbf{P}_3.$$

This ensures that the k -th column of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ correctly select the wire values from the corresponding columns $\mathbf{v}_{\mu_1(k)}$, $\mathbf{v}_{\mu_2(k)}$, and $\mathbf{v}_{\mu_3(k)}$ of $\mathbf{V}$, thereby guaranteeing that the gate operations are performed on the correct, Boolean-validated wires. Note that, since these selection matrices are sparse (each containing only s nonzero entries), the verifier can locally compute their commitments using only s MSM operations.

Matrix multiplication protocol. To prove the validity of gate permutation, we introduce an efficient matrix multiplication protocol `MatrixMul` that achieves constant-size proofs and constant verifier running time. The core idea is to reduce matrix multiplication to three matrix–vector products. For any non-zero random vector $\mathbf{r} \in \mathbb{Z}_p^\ell$, we observe the following equivalence:

$$\begin{aligned} \{\mathbf{U} = \mathbf{V}\mathbf{P}\} &\iff \{\forall \mathbf{r}, \mathbf{r}\mathbf{U} = (\mathbf{r}\mathbf{V})\mathbf{P}\} \iff \\ &\{\forall \mathbf{r}, \exists \mathbf{h}, \mathbf{d} \text{ s.t. } (\mathbf{d} = \mathbf{r}\mathbf{U}) \wedge (\mathbf{h} = \mathbf{r}\mathbf{V}) \wedge (\mathbf{d} = \mathbf{h}\mathbf{P})\}. \end{aligned}$$

Here, the intermediate vectors $\mathbf{h}$ and $\mathbf{d}$ serve as intermediate links between the three matrix–vector products, ensuring a consistent and modular proof structure.

We employ the univariate KZG polynomial commitment [30] as a succinct vector commitment and the bivariate KZG polynomial commitment [28,29,31] as a matrix commitment. Concretely, the vectors $\mathbf{r}$, $\mathbf{h}$ and $\mathbf{d}$ are interpolated as univariate polynomials $r(X)$, $h(Y)$ and $d(Z)$, respectively, while the matrices $\mathbf{P}$, $\mathbf{U}$, and $\mathbf{V}$ are represented by bivariate polynomials $P(Y, Z)$, $U(X, Z)$, and $V(X, Y)$, respectively. Thus, the commitments to $\mathbf{r}$, $\mathbf{h}$ and $\mathbf{d}$ are the univariate KZG commitments $[r(\tau_X)]$, $[h(\tau_Y)]$ and $[d(\tau_Z)]$, while the commitments to $\mathbf{P}$, $\mathbf{U}$, and $\mathbf{V}$ are the bivariate KZG evaluations $[P(\tau_Y, \tau_Z)]$, $[U(\tau_X, \tau_Z)]$, and $[V(\tau_X, \tau_Y)]$. Here, the elements

$$\{[\tau_X^i \tau_Y^j], [\tau_X^i \tau_Z^k], [\tau_Y^j \tau_Z^k]\}_{i \in [0, \ell-1], j \in [0, t-1], k \in [0, s-1]}$$

denote the trapdoor powers in the (bi)variate KZG structured reference string.

Finally, we apply the variant of bi-to-uni variate sumcheck lemma 3 to establish these three relations $\mathbf{d} = \mathbf{r}\mathbf{U}$, $\mathbf{h} = \mathbf{r}\mathbf{V}$, and $\mathbf{d} = \mathbf{h}\mathbf{P}$. These relations correspond to the following polynomial identities:

$$V(X, Y) \cdot r(X) = \frac{h(Y)}{|\mathbb{L}|} + Q(X, Y) \cdot X + D(X, Y) \cdot \zeta_{\mathbb{L}}(X),$$

$$\begin{aligned}
U(X, Z) \cdot r(X) &= \frac{d(Z)}{|\mathbb{L}|} + \bar{Q}(X, Z) \cdot X + \bar{D}(X, Z) \cdot \zeta_{\mathbb{L}}(X), \\
P(Y, Z) \cdot h(Y) &= \frac{d(Z)}{|\mathbb{V}|} + \tilde{Q}(Y, Z) \cdot Y + \tilde{D}(Y, Z) \cdot \zeta_{\mathbb{V}}(Y),
\end{aligned}$$

where $Q(X, Y), D(X, Y)$ are polynomials of degrees $\deg(X) < \ell - 1, \deg(Y) < t$, $\bar{Q}(X, Z), \bar{D}(X, Z)$ are polynomials of degrees $\deg(X) < \ell - 1, \deg(Z) < s$ and $\tilde{Q}(Y, Z), \tilde{D}(Y, Z)$ are polynomials of degrees $\deg(Y) < t - 1, \deg(Z) < s$. The proof consists of the polynomial commitments of $Q(X, Y), \bar{Q}(X, Z), \tilde{Q}(Y, Z)$ and $D(X, Y), \bar{D}(X, Z), \tilde{D}(Y, Z)$, i.e.,

$$\pi_4 = [Q(\tau_X, \tau_Y)], [\bar{Q}(\tau_X, \tau_Z)], [\tilde{Q}(\tau_Y, \tau_Z)], [D(\tau_X, \tau_Y)], [\bar{D}(\tau_X, \tau_Z)], [\tilde{D}(\tau_Y, \tau_Z)].$$

Verifying π_4 involves checking the polynomial identities above through pairing operations.

Observation. In both the bivariate zero-test relations and the matrix multiplication relations of our approach, the traditional bi-to-uni variate sumcheck lemma 3 plays a central role. However, the invocations of this bi-to-uni variate sumcheck involve multiple degree checks and divisibility checks, leading to numerous pairing-based verifications in the proof verification phase.

Improved bi-to-uni variate sumcheck. To further enhance efficiency, we extend the univariate sumcheck protocol Losum [27] to the bivariate setting, and introduce a novel non-interactive *bi-to-uni variate* sumcheck scheme, denoted BuLosum, which eliminates the need for explicit degree and divisibility checks. Our BuLosum protocol preserves the same correctness guarantees as the variant of bi-to-uni variate sumcheck 3, while achieving substantially lower proof and verification overhead, similar to Losum. Specifically, the proving phase only requires multi-scalar multiplication operations, yielding a proof represented by a single group element. Verification is likewise highly efficient, requiring only three pairing operations in total and eliminating the auxiliary degree-check and divisibility-check.

Following the same technique as in the univariate Losum [27], we note that a polynomial $F(X, Y) \cdot r(X)$ of degree at most D_X in X and $|\mathbb{V}| - 1$ in Y , with $D_X \geq |\mathbb{L}|$, satisfies

$$\sum_{\omega_L^i \in \mathbb{L}} F(\omega_L^i, Y) \cdot r(\omega_L^i) = f(Y),$$

for some polynomial $f(Y) \in \mathbb{F}^{|\mathbb{V}|-1}[Y]$ and $r(X) = \sum_{i=0}^{|\mathbb{L}|-1} \bar{r}_i X^i$, if and only if $F(X, Y) \cdot r(X) - f(Y)L_{\omega_L^*}(X)$ can be expressed as a linear combination of the following basis polynomials:

$$\begin{aligned}
&\left\{ r(\omega_L^i) \bar{L}_j(Y) L_i(X) - r(\omega_L^i) \bar{L}_j(Y) L_{i^*}(X) \mid \omega_L^i \in \mathbb{L} \setminus \{\omega_L^*\}, \omega_V^j \in \mathbb{V} \right\} \cup \\
&\left\{ \bar{r}_i X^i Y^j \zeta_{\mathbb{L}}(X) \mid i \in [0, D_X - |\mathbb{L}|], j \in [0, |\mathbb{V}| - 1] \right\}.
\end{aligned}$$

The bi-to-uni sum-check problem reduces to proving the following identity: there exist $\{F_{i,j}\}_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i*}\}, \omega_V^j \in \mathbb{V}}$ and $\{Q_{i,j}\}_{i \in [0, D_X - |\mathbb{L}|], j \in [0, |\mathbb{V}| - 1]}$ such that

$$\begin{aligned} & F(X, Y) \cdot r(X) - f(Y) L_{\omega_L^{i*}}(X) \\ &= \sum_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i*}\}} \sum_{\omega_V^j \in \mathbb{V}} F_{i,j} \cdot (r(\omega_L^i) \bar{L}_j(Y) L_i(X) - r(\omega_L^i) \bar{L}_j(Y) L_{i*}(X)) \\ &+ \sum_{i=0}^{D_X - |\mathbb{L}|} \sum_{j=0}^{|\mathbb{V}| - 1} Q_{i,j} \bar{r}_i X^i Y^j \zeta_{\mathbb{L}}(X). \end{aligned}$$

Here, ω_L^{i*} is picked randomly from $\mathbb{L}$ and the coefficients $\{Q_{i,j}\}$ correspond exactly to the quotient polynomial $Q(X, Y)$ obtained when dividing $F(X, Y) \cdot r(X)$ by $\zeta_{\mathbb{L}}(X)$. Importantly, since the prover works directly with the evaluation form of $F(X, Y)$, no additional computation is required to obtain these coefficients: indeed, $F_{i,j} = F(\omega_L^i, \omega_V^j)$.

Employing the technique established in [32,33], the setup algorithm of BuLosum initializes the CRS corresponding to the polynomial basis described above. A trusted setup party samples a random field element $\gamma \in \mathbb{Z}_p$ and precomputes the following set of group elements:

$$\begin{aligned} & \left\{ \left[\gamma \left(r(\omega_L^i) \bar{L}_j(\tau_Y) L_i(\tau_X) - r(\omega_L^i) \bar{L}_j(\tau_Y) L_{i*}(\tau_X) \right) \right] \mid \omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i*}\}, \omega_V^j \in \mathbb{V} \right\} \cup \\ & \left\{ \left[\gamma \bar{r}_i \tau_X^i \tau_Y^j \zeta_{\mathbb{L}}(\tau_X) \right] \mid i \in [0, D_X - |\mathbb{L}|], j \in [0, |\mathbb{V}| - 1] \right\}. \end{aligned}$$

All these elements are included in the CRS to enable succinct proof generation.

Given the CRS, the prover computes the proof as a linear combination of the precomputed basis elements. Specifically, for the polynomial $F(X, Y)$, the prover generates

$$\begin{aligned} \pi &= [\gamma \cdot F(\tau_X, \tau_Y) r(\tau_X)] \\ &= \sum_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i*}\}} \sum_{\omega_V^j \in \mathbb{V}} F(\omega_L^i, \omega_V^j) \cdot \left[\gamma \left(r(\omega_L^i) \bar{L}_j(\tau_Y) L_i(\tau_X) - r(\omega_L^i) \bar{L}_j(\tau_Y) L_{i*}(\tau_X) \right) \right] \\ &+ \sum_{i=0}^{D_X - |\mathbb{L}|} \sum_{j=0}^{|\mathbb{V}| - 1} Q_{i,j} \left[\gamma \bar{r}_i \tau_X^i \tau_Y^j \zeta_{\mathbb{L}}(\tau_X) \right]. \end{aligned}$$

This construction ensures that π is succinct—represented by a single group element over $\mathbb{G}_1$.

To verify the proof, the verifier checks the following pairing equation:

$$e(\pi, [\gamma^{-1}]) = e([F(\tau_X, \tau_Y)], [r(\tau_X)]) \cdot e([-f(\tau_Y)], [L_{\omega_L^{i*}}(\tau_X)]).$$

For this verification to be possible, the CRS must also include the necessary elements $[\gamma^{-1}]$ and $[r(\tau_X)]$.

Using BuLosum as a core building block, we further refine the bivariate zero-test relation and matrix multiplication relation. Based on the latter, we design an enhanced protocol MatrixMul+.

Compared to MatrixMul, the improved construction achieves both a smaller proof size and a lower verification cost, while maintaining the same level of completeness and knowledge soundness.

Organization. Section 3 introduces the necessary preliminaries, definitions, and cryptographic tools. Building upon these foundations, Section 4 presents the construction of our core building blocks. Finally, in Section 5, we detail the concrete instantiation of our batch argument and provide a comprehensive comparison with state-of-the-art BARG systems.

3 Preliminaries

Notations. For an integer $n \in \mathbb{N}$, we write $[n] := \{1, \dots, n\}$. For a finite set S , we use $s \leftarrow S$ to denote the procedure of sampling s from S uniformly. Let $\approx_s$ (resp. $\approx_c$) stand for statistical (resp. computational) indistinguishability. We use lower-case boldface to denote (column) vectors (e.g., $\mathbf{a}$) and upper-case boldface to denote matrices (e.g. $\mathbf{M}$). We use $\mathbf{a}[i]$ (resp. $\mathbf{M}[i, j]$) to indicate the entry in position i (resp. (i, j)) and use $\mathbf{a}_i$ (resp. $\mathbf{M}_i$) to denote i -th vector (resp. i -th Matrix). We write $\mathbf{a}_{1:\ell} := \{\mathbf{a}_1, \dots, \mathbf{a}_\ell\}$. For two matrices $\mathbf{A}$ and $\mathbf{B}$ of same size, the *Hadamard Product* is a matrix of the same size defined as $(\mathbf{A} \circ \mathbf{B})[i, j] = \mathbf{A}[i, j] \cdot \mathbf{B}[i, j]$. We write $\text{poly}(\lambda)$ to denote a fixed function that is bounded by some polynomial in λ and $\varepsilon(\lambda)$ to denote a function that is negligible in λ . We say that a probabilistic algorithm is PPT if it runs in polynomial time.

3.1 Bilinear Groups and Assumptions

Assume an efficient algorithm GroupGen that takes as input security parameter λ and outputs $\mathcal{G} := (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e)$. Here $\mathbb{G}_1, \mathbb{G}_2$ and $\mathbb{G}_T$ are cyclic groups of prime order p , $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ is a non-degenerate bilinear map, and all group operations and bilinear map are efficient. Let $\mathbb{G}_1 = \langle g_1 \rangle$, $\mathbb{G}_2 = \langle g_2 \rangle$ and $g_T = e(g_1, g_2)$, we employ *implicit representation* of group elements: $[x]_s = g_s^x$ and $[\mathbf{D}]_s = ([\mathbf{D}[i, j]]_s)_{ij}$ for all $s \in \{1, 2, T\}$. Given $[\mathbf{A}]_1, [\mathbf{B}]_2$, we write $e([\mathbf{A}]_1, [\mathbf{B}]_2) = [\mathbf{AB}]_T$.

Definition 1 (q -Discrete Logarithm, q -DLOG). For all $q \in \mathbb{N}$ and all PPT adversaries $\mathcal{A}_{q\text{-DLOG}}$:

$$\Pr[\tau' = \tau \mid \mathcal{G} \leftarrow \text{GroupGen}(\lambda), \tau \leftarrow \mathbb{Z}_p, \tau' \leftarrow \mathcal{A}_{q\text{-DLOG}}([1]_1, [\tau]_1, \dots, [\tau^q]_1, [1]_2, [\tau]_2)] \leq \varepsilon(\lambda).$$

3.2 Generalized Sumcheck

Polynomials and Lagrange Basis. Let $\mathbb{F}$ be a field and $\ell \in \mathbb{N}$. Let ω_L be a primitive ℓ -th root of unity that generates multiplicative subgroup $\mathbb{L} = \{\omega_L^1, \omega_L^2, \dots, \omega_L^\ell\}$. We define the set of *Lagrange*

basis polynomials $\{L_i(X)\}_{i \in [\ell]}$, where each $L_i(X)$ is the unique polynomial satisfying

$$\deg(L_i) = \ell - 1 \quad \text{and} \quad L_i(\omega_L^j) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Let $\zeta_{\mathbb{L}}(X) = \prod_{i=1}^{\ell} (X - \omega_L^i)$ be the *vanishing polynomial* on $\mathbb{L}$. We have

$$\zeta_{\mathbb{L}}(X) = X^{\ell} - 1, \quad L_i(X) = \frac{\omega_L^i}{\ell} \cdot \frac{X^{\ell} - 1}{X - \omega_L^i}, \quad L_i(0) = \frac{1}{\ell}.$$

We use bivariate polynomials to encode matrices in our scheme. Given a matrix $\mathbf{V} \in \mathbb{Z}_p^{\ell \times t}$, denote by $\mathbf{v}_j = (\mathbf{V}[1, j], \dots, \mathbf{V}[\ell, j])$ its j -th column. We encode each column vector $\mathbf{v}_j$ to a univariate polynomial

$$v_j(X) = \sum_{i \in [\ell]} \mathbf{V}[i, j] L_i(X).$$

We then aggregate all column polynomials into a bivariate polynomial

$$V(X, Y) = \sum_{j \in [t]} v_j(X) \bar{L}_j(Y)$$

where $\{\bar{L}_j(Y)\}_{j \in [t]}$ are the Lagrange basis polynomials defined over the multiplicative subgroup $\mathbb{V} = \{\omega_V^1, \dots, \omega_V^t\}$ generated by a primitive t -th root of unity ω_V . Each $\bar{L}_j(Y)$ has the explicit form

$$\bar{L}_j(Y) = \frac{\omega_V^j}{t} \cdot \frac{Y^t - 1}{Y - \omega_V^j}.$$

Lemmas. We present three straightforward variations of the univariate Sumcheck [34] known as weighted univariate Sumcheck Lemma 1, bi-to-uni Sumcheck Lemma 2 and weighted Bi-to-Uni Sumcheck Lemma 3. Among these lemmas, Lemma 3 is a combination of Lemma 1 and Lemma 2.

Lemma 1 (Weighted Univariate Sumcheck[35,36]). Denote a multiplicative subgroup $\mathbb{L} = \{\omega_L^1, \omega_L^2, \dots, \omega_L^{\ell}\}$ with $|\mathbb{L}| = \ell$. Let $L_i(X) = \frac{\omega_L^i}{|\mathbb{L}|} \cdot \frac{X^{|\mathbb{L}|-1}}{X - \omega_L^i}$, $\zeta_{\mathbb{L}}(X) = \prod_{i=1}^{|\mathbb{L}|} (X - \omega_L^i)$, $v(X) = \sum_{i=1}^{|\mathbb{L}|} v_i L_i(X)$ and $r(X) = \sum_{i=1}^{|\mathbb{L}|} r_i L_i(X)$. We have $\sum_{i=1}^{|\mathbb{L}|} v(\omega_L^i) r(\omega_L^i) = v$, if and only if there exists polynomials $Q_X(X)$ and $D_{\zeta_{\mathbb{L}}}(X)$ satisfying

$$v(X) \cdot r(X) = \frac{v}{|\mathbb{L}|} + Q_X(X) \cdot X + D_{\zeta_{\mathbb{L}}}(X) \cdot \zeta_{\mathbb{L}}(X)$$

where both $Q_X(X)$ and $D_{\zeta_{\mathbb{L}}}(X)$ are polynomials with degree $< |\mathbb{L}| - 1$ defined as

$$Q_X(X) = \sum_{i=1}^{|\mathbb{L}|} v_i r_i \frac{L_i(X) - L_i(0)}{X},$$

$$D_{\zeta_{\mathbb{L}}}(X) = \sum_{i=1}^{|\mathbb{L}|} v_i r_i \frac{L_i^2(X) - L_i(X)}{\zeta_{\mathbb{L}}(X)} + \sum_{i \neq i'} v_i r_{i'} \frac{L_{i'}(X) L_i(X)}{\zeta_{\mathbb{L}}(X)}.$$

Lemma 2 (Bi-to-Uni Sumcheck[26]). Denote two multiplicative subgroups $\mathbb{L} = \{\omega_L^1, \omega_L^2, \dots, \omega_L^\ell\}$ with $|\mathbb{L}| = \ell$, $\mathbb{V} = \{\omega_V^1, \omega_V^2, \dots, \omega_V^t\}$ with $|\mathbb{V}| = t$ and a bivariate polynomial

$$V(X, Y) = \sum_{i=1}^{|\mathbb{L}|} \sum_{j=1}^{|\mathbb{V}|} \mathbf{v}[i, j] L_i(X) \bar{L}_j(Y).$$

We have $\sum_{i=1}^{|\mathbb{L}|} V(\omega_L^i, Y) = f(Y)$, if and only if there exists a polynomial $Q(X, Y)$ satisfying

$$V(X, Y) = \frac{f(Y)}{|\mathbb{L}|} + Q(X, Y) \cdot X,$$

where $Q(X, Y)$ of $\deg(X) < |\mathbb{L}| - 1$ and $\deg(Y) < |\mathbb{V}|$ is defined as

$$Q(X, Y) = \frac{V(X, Y) - \frac{1}{|\mathbb{L}|} f(Y)}{X} = \sum_{i=1}^{|\mathbb{L}|} \sum_{j=1}^{|\mathbb{V}|} \mathbf{v}[i, j] \bar{L}_j(Y) \frac{L_i(X) - L_i(0)}{X}.$$

Lemma 3 (Weighted Bi-to-Uni Sumcheck). Let $r(X) = \sum_{i=1}^{|\mathbb{L}|} r_i L_i(X)$ and

$$V(X, Y) = \sum_{i=1}^{|\mathbb{L}|} \sum_{j=1}^{|\mathbb{V}|} \mathbf{v}[i, j] \bar{L}_j(Y) L_i(X).$$

We have $\sum_{i=1}^{|\mathbb{L}|} V(\omega_L^i, Y) \cdot r(\omega_L^i) = f(Y)$, if and only if there exist polynomials $Q_X(X, Y)$ and $D_{\zeta_L}(X, Y)$ satisfying

$$V(X, Y) \cdot r(X) = f(Y)/|\mathbb{L}| + X \cdot Q_X(X, Y) + D_{\zeta_L}(X, Y) \cdot \zeta_L(X),$$

where both $Q_X(X, Y)$ and $D_{\zeta_L}(X, Y)$ are polynomials of $\deg(X) < |\mathbb{L}| - 1$ and $\deg(Y) < |\mathbb{V}|$ defined as

$$\begin{aligned} Q_X(X, Y) &= \frac{V(X, Y) \cdot r(X) - \frac{f(Y)}{|\mathbb{L}|}}{X} = \sum_{i=1}^{|\mathbb{L}|} \sum_{j=1}^{|\mathbb{V}|} \mathbf{v}[i, j] r_i \bar{L}_j(Y) \frac{L_i(X) - L_i(0)}{X}, \\ D_{\zeta_L}(X, Y) &= \sum_{j=1}^{|\mathbb{V}|} \sum_{i=1}^{|\mathbb{L}|} \mathbf{v}[i, j] \left(r_i \bar{L}_j(Y) \frac{L_i^2(X) - L_i(X)}{\zeta_L(X)} \right) + \\ &\quad \sum_{j=1}^{|\mathbb{V}|} \sum_{i=1}^{|\mathbb{L}|} \mathbf{v}[i, j] \sum_{i'=1, i' \neq i}^{|\mathbb{L}|} \left(r_{i'} \bar{L}_j(Y) \frac{L_i(X) L_{i'}(X)}{\zeta_L(X)} \right). \end{aligned}$$

3.3 Polynomial Commitment Scheme

A Polynomial Commitment Scheme (PCS) [37] is defined by the following algorithms:

- $\text{KeyGen}(\lambda, d, m) \rightarrow \text{par}$: It takes as input a security number $\lambda \in \mathbb{N}$, variate-degree bound $d \in \mathbb{N}$ and variate $m \in \mathbb{N}$, generates public parameter par .

- $\text{Commit}(\text{par}, f) \rightarrow \text{com}_f$: It takes as input a polynomial $f(\mathbf{X})$ where $\mathbf{X} = (X_0, \dots, X_{m-1})$ and par , outputs commitment com_f .
- $\text{Open}(\text{par}, \text{com}_f, \mathbf{x}) \rightarrow (\pi_f, z)$: It takes as input par, com_f and the evaluation point $\mathbf{x}$, generates a proof π_f and the value $z = f(\mathbf{x})$.
- $\text{Verify}(\text{par}, \text{com}_f, \mathbf{x}, \pi_f, z) \rightarrow b$: It is a protocol between the prover and the verifier. The verifier checks whether $z = f(\mathbf{x})$ through input $\text{par}, \text{com}_f, \mathbf{x}, \pi_f$ and outputs $b \in \{0, 1\}$.

This polynomial commitment scheme satisfies the following properties:

Completeness. For all $\lambda, d, m \in \mathbb{N}$, all evaluation points $\mathbf{x} \in \mathbb{Z}_p^m$, and any polynomial f with degree bound d and variate m , it holds that

$$\Pr \left[\text{Verify}(\text{par}, \text{com}_f, \mathbf{x}, \pi_f, z) = 1 \mid \begin{array}{l} \text{par} \leftarrow \text{KeyGen}(\lambda, d, m); \\ \text{com}_f \leftarrow \text{Commit}(\text{par}, f); \\ (\pi_f, z) \leftarrow \text{Open}(\text{par}, \text{com}_f, \mathbf{x}) \end{array} \right] = 1.$$

Knowledge soundness. For every PPT adversary $\mathcal{A}_{\text{PCS}}$, there exists a PPT extractor $\mathcal{E}_{\text{PCS}}$ with access to $\mathcal{A}_{\text{PCS}}$'s messages during the protocol, it holds that

$$\Pr \left[\begin{array}{l} \text{Verify}(\text{par}, \text{com}, \mathbf{x}, \pi, z) = 1 \wedge \\ \text{com} = \text{Commit}(\text{par}, f) \wedge \\ f(\mathbf{x}) \neq z \end{array} \mid \begin{array}{l} \text{par} \leftarrow \text{KeyGen}(\lambda, d, m); \\ (z, \mathbf{x}) \leftarrow \mathcal{A}_{\text{PCS}}(\text{par}, \lambda); \\ (\text{com}, \pi) \leftarrow \mathcal{A}_{\text{PCS}}(\text{par}, \lambda); \\ f \leftarrow \mathcal{E}_{\text{PCS}}^{\mathcal{A}_{\text{PCS}}(\cdot)}(\text{par}, \lambda) \end{array} \right] \leq \varepsilon(\lambda).$$

3.4 Algebraic Group Model

The *Algebraic Group Model* (AGM) [38] is a framework for analyzing the security of cryptographic schemes instantiated over algebraic groups. In this model, any PPT adversary $\mathcal{A}_{\text{AGM}}$ operating in a bilinear group is restricted to producing new group elements only via explicit algebraic combinations of elements it has previously observed. Formally, let $\mathbb{G}_1$ and $\mathbb{G}_2$ be finite cyclic groups of prime order p , and let $h_1, \dots, h_n \in \mathbb{G}_i$ (for $i \in \{1, 2\}$) denote the group elements that have appeared so far in the execution. Every PPT adversary $\mathcal{A}_{\text{AGM}}$ that outputs a group element $h \in \mathbb{G}_i$ is required to also output its *algebraic representation*, i.e., a vector $(a_1, \dots, a_n) \in \mathbb{Z}_p^n$ such that

$$h = h_1^{a_1} \cdots h_n^{a_n}.$$

This vector $(a_1, \dots, a_n)$ is referred to as the *algebraic coefficients* of h . This requirement captures the constraint that $\mathcal{A}_{\text{AGM}}$ can only generate new group elements as algebraic combinations of previously known ones. Such an adversary is called *algebraic*.

3.5 Batch Arguments

A *non-interactive Batch Argument* (BARG) [23,20] for circuit satisfiability is defined by the following algorithms:

- $\text{Setup}(\lambda, \ell, s) \rightarrow \text{crs}$: It takes as input a security number $\lambda \in \mathbb{N}$, the number of instances $\ell \in \mathbb{N}$ and upper bound of circuit size $s \in \mathbb{N}$, outputs a common reference string crs .
- $\text{Prove}(\text{crs}, C, (\mathbf{x}_1, \dots, \mathbf{x}_\ell), (\mathbf{w}_1, \dots, \mathbf{w}_\ell)) \rightarrow \pi$: It takes as input crs , a Boolean circuit $C : \{0, 1\}^n \times \{0, 1\}^h \rightarrow \{0, 1\}$, statements $\mathbf{x}_1, \dots, \mathbf{x}_\ell \in \{0, 1\}^n$ and witnesses $\mathbf{w}_1, \dots, \mathbf{w}_\ell \in \{0, 1\}^h$, outputs a proof π .
- $\text{Verify}(\text{crs}, C, (\mathbf{x}_1, \dots, \mathbf{x}_\ell), \pi) \rightarrow b$: It takes as input crs , $C : \{0, 1\}^n \times \{0, 1\}^h \rightarrow \{0, 1\}$, $\mathbf{x}_1, \dots, \mathbf{x}_\ell \in \{0, 1\}^n$ and a proof π , outputs $b \in \{0, 1\}$.

Here, we require that Verify is deterministic and the batch argument satisfies the following properties:

Completeness. For all $\lambda, \ell, s \in \mathbb{N}$, all Boolean circuits $C : \{0, 1\}^n \times \{0, 1\}^h \rightarrow \{0, 1\}$ of size at most $s(\lambda)$, all statements $\mathbf{x}_1, \dots, \mathbf{x}_\ell \in \{0, 1\}^n$, and all witnesses $\mathbf{w}_1, \dots, \mathbf{w}_\ell \in \{0, 1\}^h$ satisfying $C(\mathbf{x}_i, \mathbf{w}_i) = 1$ for all $i \in [\ell]$, it holds that

$$\Pr \left[\text{Verify}(\text{crs}, C, \mathbf{x}_{1:\ell}, \pi) = 1 \mid \begin{array}{l} \text{crs} \leftarrow \text{Setup}(\lambda, \ell, s); \\ \pi \leftarrow \text{Prove}(\text{crs}, C, \mathbf{x}_{1:\ell}, \mathbf{w}_{1:\ell}) \end{array} \right] = 1.$$

Non-adaptive soundness. For every PPT adversary $\mathcal{A}_{\text{BARG}}$, all Boolean circuits $C : \{0, 1\}^n \times \{0, 1\}^h \rightarrow \{0, 1\}$ of size at most $s(\lambda)$, and all statements $\mathbf{x}_1, \dots, \mathbf{x}_\ell$, if there exists some $i \in [\ell]$ such that $C(\mathbf{x}_i, \mathbf{w}_i) \neq 1$, then

$$\Pr \left[\text{Verify}(\text{crs}, C, \mathbf{x}_{1:\ell}, \pi) = 1 \mid \begin{array}{l} \text{crs} \leftarrow \text{Setup}(\lambda, \ell, s); \\ \pi \leftarrow \mathcal{A}_{\text{BARG}}(\text{crs}, C, \mathbf{x}_{1:\ell}) \end{array} \right] \leq \varepsilon(\lambda).$$

4 Building Blocks

In this section, we introduce a novel non-interactive sum-check scheme, denoted BuLosum. Then, we use BuLosum to construct an efficient matrix multiplication protocol MatrixMul+.

4.1 Optimal Bi-to-uni variate Sumcheck

We present the optimal bi-to-uni variate sumcheck from the framework of [27]. Its proving cost is merely a single multi-scalar multiplication, yielding a proof that is just one group element. Verification is dominated by only three pairings.

Protocol description. A formal description of this protocol is provided below.

- **Setup**($\mathbb{V}, \mathbb{L}, \mathcal{G}, \text{crs}_{\text{KZG}}$) : On input subgroups $\mathbb{V} = \{\omega_V, \dots, \omega_V^t\}$ and $\mathbb{L} = \{\omega_L, \dots, \omega_L^\ell\}$ of sizes t and ℓ respectively, bilinear pairing parameters $\mathcal{G} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e) \leftarrow \text{GroupGen}(1^\lambda)$, and a bivariate KZG CRS $\text{crs}_{\text{KZG}} = \left(\{[\tau_X^i \tau_Y^j]_1\}_{i \in [0, D_X-1], j \in [0, t-1]}, [\tau_X \tau_Y]_2 \right)$ where $D_X \geq |\mathbb{L}|$, the algorithm samples $\gamma \xleftarrow{\$} \mathbb{Z}_p$ and a random vector $\mathbf{r} = (r_1, \dots, r_\ell) \in \mathbb{Z}_p^\ell$ defining the polynomial $r(X) = \sum_{i \in [\ell]} r_i L_i(X)$. Rewrite $r(X)$ as $\sum_{i=0}^{|\mathbb{L}|-1} \bar{r}_i X^i$. It then outputs the CRS:

$$\text{crs} = \left(\begin{array}{l} \left\{ \left[\gamma \cdot \left(r_i L_i(\tau_X) \bar{L}_j(\tau_Y) - r_i \bar{L}_j(\tau_Y) L_{i^*}(\tau_X) \right) \right]_1 \right\}_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i^*}\}, \omega_V^j \in \mathbb{V}}, \\ \left\{ \left[\gamma \cdot \bar{r}_i \tau_X^i \tau_Y^j \zeta_{\mathbb{L}}(\tau_X) \right]_1 \right\}_{i \in [0, D_X-\ell], j \in [0, t-1]}, \\ [1]_1, [L_{i^*}(\tau_X)]_1, [\gamma^{-1}]_2, [r(\tau_X)]_2 \end{array} \right)$$

where $\omega_L^{i^*}$ is picked from $\mathbb{L}$ arbitrarily.

- **Prove**($\text{crs}, \mathbb{L}, \mathbb{V}, F(X, Y)$) : On input $F(X, Y)$, $\mathbb{L}, \mathbb{V}$ and crs , the algorithm proceeds as follows:
 1. Divide $F(X, Y) \cdot r(X)$ by $\zeta_{\mathbb{L}}(X)$ and let the quotient be $Q(X, Y) = \sum_{i=0}^{D_X-|\mathbb{L}|} \sum_{j=0}^{|\mathbb{V}|-1} Q_{i,j} \bar{r}_i X^i Y^j$.
 2. Output the proof

$$\begin{aligned} \pi = & \sum_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i^*}\}} \sum_{\omega_V^j \in \mathbb{V}} F(\omega_L^i, \omega_V^j) \cdot \left[\gamma \left(r_i \bar{L}_j(\tau_Y) L_i(\tau_X) - r_i \bar{L}_j(\tau_Y) L_{i^*}(\tau_X) \right) \right]_1 \\ & + \sum_{i=0}^{D_X-|\mathbb{L}|} \sum_{j=0}^{|\mathbb{V}|-1} Q_{i,j} \left[\gamma \bar{r}_i \tau_X^i \tau_Y^j \zeta_{\mathbb{L}}(\tau_X) \right]_1. \end{aligned}$$

- **Verify**($\text{crs}, \pi, [F(\tau_X, \tau_Y)]_1, [f(\tau_Y)]_1$) : On input $\text{crs}, \pi, [F(\tau_X, \tau_Y)]_1, [f(\tau_X)]_1$, the algorithm checks

$$e([F(\tau_X, \tau_Y)]_1, [r(\tau_X)]_2) \cdot e([L_{i^*}(\tau_X)]_1, -[f(\tau_Y)]_2) = e(\pi, [\gamma^{-1}]_2).$$

Completeness If both the prover and the verifier follow the protocol honestly, then all verification equations hold by construction. Let

$$\tilde{F}(X, Y) \cdot \tilde{r}(X) := F(X, Y) \cdot r(X) \bmod \zeta_{\mathbb{L}}(X).$$

Then the right-hand side of the verification equation, after division by γ , has exponent

$$\begin{aligned}
& \sum_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i*}\}} \sum_{\omega_V^j \in \mathbb{V}} F(\omega_L^i, \omega_V^j) \cdot (r_i \bar{L}_j(\tau_Y) L_i(\tau_X) - r_i \bar{L}_j(\tau_Y) L_{i^*}(\tau_X)) + Q(\tau_X, \tau_Y) \zeta_{\mathbb{L}}(\tau_X) \\
&= \sum_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i*}\}} \sum_{\omega_V^j \in \mathbb{V}} F(\omega_L^i, \omega_V^j) \cdot r_i \bar{L}_j(\tau_Y) L_i(\tau_X) - \sum_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i*}\}} \sum_{\omega_V^j \in \mathbb{V}} F(\omega_L^i, \omega_V^j) \cdot r_i \bar{L}_j(\tau_Y) L_{i^*}(\tau_X) \\
&\quad + Q(\tau_X, \tau_Y) \zeta_{\mathbb{L}}(\tau_X) \\
&= \sum_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i*}\}} F(\omega_L^i, \tau_Y) \cdot r_i L_i(\tau_X) - \sum_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i*}\}} F(\omega_L^i, \tau_Y) \cdot r_i L_{i^*}(\tau_X) \\
&\quad + Q(\tau_X, \tau_Y) \zeta_{\mathbb{L}}(\tau_X) \\
&= \sum_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i*}\}} F(\omega_L^i, \tau_Y) \cdot r(\omega_L^i) L_i(\tau_X) - (f(\tau_Y) - F(\omega_L^{i*}, \tau_Y) \cdot r(\omega_L^{i*})) L_{i^*}(\tau_X) \\
&\quad + Q(\tau_X, \tau_Y) \zeta_{\mathbb{L}}(\tau_X) \\
&= \sum_{\omega_L^i \in \mathbb{L}} F(\omega_L^i, \tau_Y) \cdot r(\omega_L^i) L_i(\tau_X) - f(\tau_Y) L_{i^*}(\tau_X) + Q(\tau_X, \tau_Y) \zeta_{\mathbb{L}}(\tau_X) \\
&= \tilde{F}(\tau_X, \tau_Y) \cdot \tilde{r}(\tau_X) - f(\tau_Y) \cdot L_{i^*}(\tau_X) + Q(\tau_X, \tau_Y) \zeta_{\mathbb{L}}(\tau_X) \\
&= F(\tau_X, \tau_Y) \cdot r(\tau_X) - f(\tau_Y) \cdot L_{i^*}(\tau_X).
\end{aligned}$$

This matches the exponent of the left-hand side, and hence the protocol satisfies perfect completeness. $\square$

Security We prove the security of BuLosum in the algebraic group model. In more detail, we will prove the following theorem.

Theorem 1 (Knowledge Soundness). *The introduced bi-to-uni variate sumcheck BuLosum achieves knowledge soundness in the algebraic group model.*

Proof. Since $\mathcal{A}_{\text{BuLosum}}$ is algebraic, whenever $\mathcal{A}_{\text{BuLosum}}$ outputs $\text{com}_F, \text{com}_f, \pi \in \mathbb{G}_1$, it can simultaneously provide the corresponding linear coefficients with respect to the CRS. Hence, we can construct an extractor $\mathcal{E}$ that runs the same strategy as $\mathcal{A}_{\text{BuLosum}}$, retrieves these coefficients, and computes the polynomials $F_1(X, Y), F_2(X, Y), P_1(X, Y), P_2(X, Y)$ such that

$$\text{com}_F = [F_1(\tau_X, \tau_Y) + \gamma F_2(\tau_X, \tau_Y)]_1, \quad \pi = [P_1(\tau_X, \tau_Y)]_1 + \gamma [P_2(\tau_X, \tau_Y)]_1.$$

If the verification passes, i.e.,

$$e([F_1(\tau_X, \tau_Y) + \gamma F_2(\tau_X, \tau_Y)]_1, [r(\tau_X)]_2) \cdot e([f(\tau_Y)]_1, -[L_{i^*}(\tau_X)]_2) = e(\pi, [\gamma^{-1}]_2),$$

then we obtain

$$\begin{aligned}
& \gamma(F_1(\tau_X, \tau_Y) \cdot r(\tau_X) - f(\tau_Y) L_{i^*}(\tau_X)) + \gamma^2 F_2(\tau_X, \tau_Y) r(\tau_X) \\
&= P_1(\tau_X, \tau_Y) + \gamma P_2(\tau_X, \tau_Y).
\end{aligned}$$

This can be rewritten as

$$-P_1(\tau_X, \tau_Y) + \gamma(F_1(\tau_X, \tau_Y) \cdot r(\tau_X) - f(\tau_Y)L_{i^*}(\tau_X) - P_2(\tau_X, \tau_Y)) + \gamma^2 F_2(\tau_X, \tau_Y)r(\tau_X) = 0.$$

Since γ is sampled independently of the adversary, the above identity implies that each coefficient must vanish individually; otherwise, one would obtain a non-trivial polynomial relation in γ and would allow extracting γ from $[\gamma]_1$ and $[\gamma^{-1}]_2$, breaking the q -DLOG assumption. Therefore, we conclude that

$$\begin{aligned} P_1(\tau_X, \tau_Y) &= 0, F_1(\tau_X, \tau_Y) \cdot r(\tau_X) - f(\tau_Y)L_{i^*}(\tau_X) - P_2(\tau_X, \tau_Y) = 0, \\ F_2(\tau_X, \tau_Y)r(\tau_X) &= 0. \end{aligned}$$

By standard KZG knowledge soundness arguments, these relations must hold for all $X \in \mathbb{L}$ and $Y \in \mathbb{V}$. Otherwise, one could build an adversary that solves for $\tau_X \tau_Y$ from the KZG parameters. Consequently, we obtain

$$F(X, Y) \cdot r(X) = F_1(X, Y) \cdot r(X) = f(Y)L_{i^*}(X) + P_2(X, Y),$$

where $P_2(X, Y)$ is a linear combination of terms of the form

$$r_i L_i(X) \bar{L}_j(Y) - r_i \bar{L}_j(Y) L_{i^*}(X) \quad \text{and} \quad \bar{r}_i X^i Y^j \zeta_{\mathbb{L}}(X).$$

Finally, evaluating over the subgroup $\mathbb{L}$ yields

$$\sum_{\omega_L^i \in \mathbb{L}} F(\omega_L^i, Y) \cdot r(\omega_L^i) = f(Y).$$

□

4.2 MatrixMul+: Improved Matrix Multiplication Protocol

In this subsection, we present a more efficient matrix multiplication protocol, MatrixMul+, which reduces both proof size and verification cost compared to MatrixMul in Appendix A. The protocol follows the same high-level framework as MatrixMul, namely reducing matrix multiplication to three matrix-vector products. However, instead of using the variant of bi-to-uni variate sum-check [26], we employ the introduced BuLosum in Subsection 4.1 to prove these three matrix-vector product relations.

Protocol description. A formal description of the improved matrix multiplication argument MatrixMul+, is provided below.

- Setup(λ, ℓ, t, s) $\rightarrow$ crs:
 1. Run $\mathcal{G} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e) \leftarrow \text{GroupGen}(\lambda)$.

2. Denote three multiplicative subgroups $\mathbb{L} = \{\omega_L, \dots, \omega_L^\ell\}$, $\mathbb{V} = \{\omega_V, \dots, \omega_V^t\}$, and $\mathbb{H} = \{\omega_H, \dots, \omega_H^s\}$ such that $|\mathbb{L}| = \ell$, $|\mathbb{V}| = t$, $|\mathbb{H}| = s$.
3. Define $L_i(X) = \frac{\omega_L^i}{|\mathbb{L}|} \cdot \frac{X^{|\mathbb{L}|-1}}{X - \omega_L^i}$, $\bar{L}_j(Y) = \frac{\omega_V^j}{|\mathbb{V}|} \cdot \frac{Y^{|\mathbb{V}|-1}}{Y - \omega_V^j}$ and $\tilde{L}_k(Z) = \frac{\omega_H^k}{|\mathbb{H}|} \cdot \frac{Z^{|\mathbb{H}|-1}}{Z - \omega_H^k}$.
4. Sample $\gamma, \tau_X, \tau_Y, \tau_Z \leftarrow \mathbb{Z}_p$. Sample a random vector $\mathbf{r} = (r_1, \dots, r_\ell) \in \mathbb{Z}_p^\ell$ defining the polynomial $r(X) = \sum_{i \in [\ell]} r_i L_i(X)$. Rewrite $r(X)$ as $\sum_{i=0}^{|\mathbb{L}|-1} \bar{r}_i X^i$. Output crs as

$$\left(\begin{array}{l} \left\{ \left[\gamma \cdot \left(r_i L_i(\tau_X) \bar{L}_j(\tau_Y) - r_i \bar{L}_j(\tau_Y) L_{i^*}(\tau_X) \right) \right]_1 \right\}_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i^*}\}, \omega_V^j \in \mathbb{V}}, \\ \left\{ \left[\gamma \cdot \left(r_i L_i(\tau_X) \tilde{L}_k(\tau_Z) - r_i \tilde{L}_k(\tau_Z) L_{i^*}(\tau_X) \right) \right]_1 \right\}_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i^*}\}, \omega_H^k \in \mathbb{H}}, \\ \left\{ \left[\gamma \cdot \left(r_i \tilde{L}_k(\tau_Z) \bar{L}_j(\tau_Y) - r_i \tilde{L}_k(\tau_Z) \bar{L}_{j^*}(\tau_Y) \right) \right]_1 \right\}_{\omega_L^j \in \mathbb{V} \setminus \{\omega_V^{j^*}\}, \omega_H^k \in \mathbb{H}}, \\ \left\{ \left[\gamma \cdot \bar{r}_i \tau_X^i \tau_Y^j \zeta_{\mathbb{L}}(\tau_X) \right]_1 \right\}_{i \in [0, D_X - \ell], j \in [0, t-1]}, \\ \left\{ \left[\gamma \bar{r}_i \tau_X^i \tau_Z^k \zeta_{\mathbb{L}}(\tau_X) \right]_1 \right\}_{i \in [0, D_X - \ell], k \in [0, s-1]}, \\ \left\{ \left[\gamma \bar{r}_i \tau_Y^j \tau_Z^k \zeta_{\mathbb{V}}(\tau_Y) \right]_1 \right\}_{i \in [0, \ell-1], j \in [0, D_Y - t], k \in [0, s-1]}, \\ \left\{ [\tau_X^i \tau_Y^j]_1, [\tau_X^i \tau_Z^k]_1, [\tau_Y^j \tau_Z^k]_1 \right\}_{i \in [0, \ell-1], j \in [0, t-1], k \in [0, s-1]}, \\ \left\{ [r_i]_1, [r_i L_i(\tau_X)]_2, [r_i \bar{L}_j(\tau_Y)]_2, [r_i \tilde{L}_k(\tau_Z)]_2 \right\}_{i \in [\ell], j \in [t], k \in [s]}, \\ [1]_1, \mathbb{G}, [L_{i^*}(\tau_X)]_1, [\bar{L}_{j^*}(\tau_Y)]_1, [\gamma^{-1}]_2, [r(\tau_X)]_2. \end{array} \right)$$

– Prove(crs, $\mathbf{P}, \mathbf{U}, \mathbf{V}$) : Take as input crs and three matrices $\mathbf{P} \in \mathbb{Z}_p^{t \times s}$, $\mathbf{U} \in \mathbb{Z}_p^{\ell \times s}$, $\mathbf{V} \in \mathbb{Z}_p^{\ell \times t}$, then proceed as follows:

1. Compute three polynomials $P(Y, Z)$, $V(X, Y)$ and $U(X, Z)$ by interpolating $\mathbf{P}[j, k]$ on all (ω_V^j, ω_H^k) , $\mathbf{V}[i, j]$ on all (ω_L^i, ω_V^j) and $\mathbf{U}[i, k]$ on all (ω_L^i, ω_H^k) , respectively. Set

$$P = \left[\sum_{k=1}^s \sum_{j=1}^t \mathbf{P}[j, k] \bar{L}_j(\tau_Y) \tilde{L}_k(\tau_Z) \right]_1, \quad V = \left[\sum_{j=1}^t \sum_{i=1}^\ell \mathbf{V}[i, j] L_i(\tau_X) \bar{L}_j(\tau_Y) \right]_1,$$

$$U = \left[\sum_{k=1}^s \sum_{i=1}^\ell \mathbf{U}[i, k] L_i(\tau_X) \tilde{L}_k(\tau_Z) \right]_1.$$

2. Set the intermediate vectors $\mathbf{d}, \mathbf{h}$ as

$$[d(\tau_Z)]_2 = \sum_{i=1}^\ell \sum_{k=1}^s U(\omega_L^i, \omega_H^k) [r_i \tilde{L}_k(\tau_Z)]_2,$$

$$[h(\tau_Y)]_2 = \sum_{i=1}^\ell \sum_{j=1}^t V(\omega_L^i, \omega_V^j) [r_i \bar{L}_j(\tau_Y)]_2.$$

3. Divide $V(X, Y) \cdot r(X)$ by $\zeta_{\mathbb{L}}(X)$, $U(X, Y) \cdot r(X)$ by $\zeta_{\mathbb{L}}(X)$ and $P(Y, Z) \cdot h(Y)$ by $\zeta_{\mathbb{V}}(Y)$. Let $Q(X, Y) = \sum_{i=0}^{D_X-|\mathbb{L}|} \sum_{j=0}^{|\mathbb{V}|-1} Q_{i,j} \bar{r}_i X^i Y^j$, $\bar{Q}(X, Z) = \sum_{i=0}^{D_X-|\mathbb{L}|} \sum_{k=0}^{|\mathbb{H}|-1} \bar{Q}_{i,k} \bar{r}_i X^i Z^k$, $\tilde{Q}(Y, Z) = \sum_{j=0}^{D_Y-|\mathbb{V}|} \sum_{k=0}^{|\mathbb{H}|-1} \sum_{i=0}^{|\mathbb{L}|-1} \tilde{Q}_{j,k} \bar{r}_i Y^j Z^k$ be the corresponding quotients respectively. Output the proof $\pi = (\pi_1, \pi_2, \pi_3)$ where

$$\begin{aligned} \pi_1 &= \sum_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i^*}\}} \sum_{\omega_V^j \in \mathbb{V}} V(\omega_L^i, \omega_V^j) \cdot \left[\gamma \left(r_i \bar{L}_j(\tau_Y) L_i(\tau_X) - r_i \bar{L}_j(\tau_Y) L_{i^*}(\tau_X) \right) \right]_1 \\ &\quad + \sum_{i=0}^{D_X-|\mathbb{L}|} \sum_{j=0}^{|\mathbb{V}|-1} Q_{i,j} \left[\gamma \bar{r}_i \tau_X^i \tau_Y^j \zeta_{\mathbb{L}}(\tau_X) \right]_1, \\ \pi_2 &= \sum_{\omega_L^i \in \mathbb{L} \setminus \{\omega_L^{i^*}\}} \sum_{\omega_H^k \in \mathbb{H}} U(\omega_L^i, \omega_H^k) \cdot \left[\gamma \left(r_i \tilde{L}_k(\tau_Z) L_i(\tau_X) - r_i \tilde{L}_k(\tau_Z) L_{i^*}(\tau_X) \right) \right]_1 \\ &\quad + \sum_{i=0}^{D_X-|\mathbb{L}|} \sum_{k=0}^{|\mathbb{H}|-1} \bar{Q}_{i,k} \left[\gamma \bar{r}_i \tau_X^i \tau_Z^k \zeta_{\mathbb{L}}(\tau_X) \right]_1, \\ \pi_3 &= \sum_{\omega_V^j \in \mathbb{L} \setminus \{\omega_V^{j^*}\}} \sum_{\omega_H^k \in \mathbb{H}} P(\omega_V^j, \omega_H^k) \cdot \sum_{i=1}^{|\mathbb{L}|} V(\omega_L^i, \omega_V^j) \left[\gamma \left(r_i \tilde{L}_k(\tau_Z) \bar{L}_j(\tau_Y) - \right. \right. \\ &\quad \left. \left. r_i \tilde{L}_k(\tau_Z) \bar{L}_{j^*}(\tau_Y) \right) \right]_1 + \sum_{j=0}^{D_Y-|\mathbb{V}|} \sum_{k=0}^{|\mathbb{H}|-1} \tilde{Q}_{j,k} \sum_{i=0}^{|\mathbb{L}|-1} \left[\gamma \bar{r}_i \tau_Y^j \tau_Z^k \zeta_{\mathbb{V}}(\tau_Y) \right]_1. \end{aligned}$$

– Verify(crs, π , P , V , U) $\rightarrow \{0, 1\}$:

1. Parse $\pi = (\pi_1, \pi_2, \pi_3)$.
2. Verify bilinear pairing relation and output 1 iff

$$\begin{aligned} e(V, [r(\tau_X)]_2) \cdot e([L_{i^*}(\tau_X)]_1, -[h(\tau_Y)]_2) &= e(\pi_1, [\gamma^{-1}]_2) \wedge \\ e(U, [r(\tau_X)]_2) \cdot e([L_{i^*}(\tau_X)]_1, -[d(\tau_Z)]_2) &= e(\pi_2, [\gamma^{-1}]_2) \wedge \\ e(P, [h(\tau_Y)]_2) \cdot e([\bar{L}_{j^*}(\tau_Y)]_1, -[d(\tau_Z)]_2) &= e(\pi_3, [\gamma^{-1}]_2). \end{aligned}$$

Completeness If the underlying BuLosum protocol is perfectly complete, then the composed scheme MatrixMul+ is also perfectly complete. An honest prover holding matrices $\mathbf{P}, \mathbf{U}, \mathbf{V}$ with $\mathbf{V} \cdot \mathbf{P} = \mathbf{U}$ produces proofs (π_1, π_2, π_3) that each pass their respective BuLosum check. Since the verifier accepts iff all three checks succeed, the composed protocol is perfectly complete.

Security We now argue that the security of our proposed scheme MatrixMul+ can be reduced to the security of the underlying protocol BuLosum. In more detail, we will prove the following theorem.

Theorem 2 (Knowledge Soundness). *If the Bi-to-uni variate sumcheck protocol BuLosum is knowledge-sound in the algebraic group model, then the proposed scheme also achieves knowledge soundness in the algebraic group model.*

Proof (Proof Sketch). At a high level, the proof follows by reduction. Assume, towards contradiction, that there exists an adversary $\mathcal{A}_{\text{MatrixMul+}}$ that can break the knowledge soundness of our scheme with non-negligible probability. That is, $\mathcal{A}_{\text{MatrixMul+}}$ can produce a convincing proof $\pi = (\pi_1, \pi_2, \pi_3)$ together with commitments P, V, U that pass verification, without knowledge of valid witnesses $(\mathbf{P}, \mathbf{U}, \mathbf{V})$ satisfying $\mathbf{V} \cdot \mathbf{P} = \mathbf{U}$.

We now construct an adversary $\mathcal{B}_{\text{BuLosum}}$ against the bi-to-uni variate sumcheck protocol BuLosum. $\mathcal{B}_{\text{BuLosum}}$ internally simulates the introduced scheme for $\mathcal{A}_{\text{MatrixMul+}}$ by embedding the BuLosum CRS into the CRS of our protocol. In particular, each of the three consistency checks of our scheme (for V, U , and P) is realized by an independent instantiation of BuLosum on the corresponding polynomial identity:

$$\begin{aligned} V(X, Y) \cdot r(X) &\equiv \sum_i \mathbf{V}[i, \cdot] \cdot r_i L_i(X), & U(X, Z) \cdot r(X) &\equiv \sum_i \mathbf{U}[i, \cdot] \cdot r_i L_i(X), \\ P(Y, Z) \cdot h(Y) &\equiv \sum_j \mathbf{P}[j, \cdot] \cdot h_j \bar{L}_j(Y). \end{aligned}$$

If $\mathcal{A}_{\text{MatrixMul+}}$ outputs a valid proof π that causes the verifier to accept, then at least one of the three BuLosum instances has been solved without a corresponding witness. In this case, $\mathcal{B}_{\text{BuLosum}}$ extracts a violating proof for the underlying protocol BuLosum. By the knowledge soundness of BuLosum (Theorem 4.1), this is impossible except with negligible probability.

Therefore, $\mathcal{A}_{\text{MatrixMul+}}$ cannot succeed without actually knowing valid witnesses for $\mathbf{P}, \mathbf{U}, \mathbf{V}$, and thus our scheme inherits knowledge soundness from the underlying bi-to-uni variate sumcheck protocol BuLosum. $\square$

5 Efficient Pairing-Based Batch Arguments for NP

In this section, we show how to construct an efficient BARG with constant-size proof and constant number of pairing operations. Our construction builds on the improved bivariate zero-test [26] (realized via our introduced BuLosum in Section 4.1) together with our proposed MatrixMul+ protocol in Section 4.2.

5.1 Protocol Description

We now construct our BARG for the language of circuit satisfiability as follows.

- $\text{Setup}(\lambda, \ell, t, s) \rightarrow \text{crs}$:
 1. Run $\mathcal{G} := (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e) \leftarrow \text{GroupGen}(\lambda)$.
 2. Denote three multiplicative subgroups $\mathbb{L} = \{\omega_L, \dots, \omega_L^\ell\}$, $\mathbb{V} = \{\omega_V, \dots, \omega_V^t\}$, $\mathbb{H} = \{\omega_H, \dots, \omega_H^s\}$ such that $|\mathbb{L}| = \ell$, $|\mathbb{V}| = t$, $|\mathbb{H}| = s$.

3. For each $i \in [\ell], j \in [t], k \in [s]$, define the Lagrange basis polynomials $L_i(X) = \frac{\omega_L^i}{|\mathbb{L}|} \cdot \frac{X^{|\mathbb{L}|-1}}{X - \omega_L^i}$, $\bar{L}_j(Y) = \frac{\omega_V^j}{|\mathbb{V}|} \cdot \frac{Y^{|\mathbb{V}|-1}}{Y - \omega_V^j}$, $\tilde{L}_k(Z) = \frac{\omega_H^k}{|\mathbb{H}|} \cdot \frac{Z^{|\mathbb{H}|-1}}{Z - \omega_H^k}$, and the corresponding vanishing polynomials $\zeta_{\mathbb{L}}(X) = \prod_{i \in [\ell]} (X - \omega_L^i)$, $\zeta_{\mathbb{V}}(Y) = \prod_{j \in [t]} (Y - \omega_V^j)$, $\zeta_{\mathbb{H}}(Z) = \prod_{k \in [s]} (Z - \omega_H^k)$.
4. Define auxiliary linearly independent polynomials $\rho_j(R), \bar{\rho}_k(R)$ for $j \in [t], k \in [s]$. Sample $\tau_X, \tau_Y, \tau_Z, \tau_R \leftarrow \mathbb{Z}_p$, the auxiliary string references

$$\text{ars} = \left(\left\{ \left[L_i(\tau_X) \bar{L}_j(\tau_Y), L_i(\tau_X) \tilde{L}_k(\tau_Z), \bar{L}_j(\tau_Y) \tilde{L}_k(\tau_Z) \right]_{1,2} \right\}_{i \in [\ell], j \in [t], k \in [s]} \right. \\ \left. \left\{ \left[\rho_j(\tau_R) \bar{L}_j(\tau_Y), \bar{\rho}_k(\tau_R) \tilde{L}_k(\tau_Z) \right]_2 \right\}_{j \in [t], k \in [s]} \right. \\ \left. \left\{ \left[\tau_X^{i-1} \tau_Y^{j-1}, \tau_X^{i-1} \tau_Z^{k-1}, \tau_Y^{j-1} \tau_Z^{k-1} \right]_{1,2} \right\}_{i \in [\ell], j \in [t], k \in [s]} \right)$$

5. Run to generate BuLosum CRS $\text{crs}_{\text{BuLosum}}^{(1)} \leftarrow \text{BuLosum.Setup}(\mathcal{G}, \mathbb{L}, \mathbb{V}, \text{ars})$, $\text{crs}_{\text{BuLosum}}^{(2)} \leftarrow \text{BuLosum.Setup}(\mathcal{G}, \mathbb{L}, \mathbb{H}, \text{ars})$.
 6. Run $\text{crs}_{\text{MatrixMul+}} \leftarrow \text{MatrixMul+.Setup}(\lambda, \ell, t, s)$.
 7. Output $\text{crs} = (\text{ars}, \mathcal{G}, \text{crs}_{\text{BuLosum}}^{(1)}, \text{crs}_{\text{BuLosum}}^{(2)}, \text{crs}_{\text{MatrixMul+}})$.
- Prove($\text{crs}, C, (\mathbf{x}_1, \dots, \mathbf{x}_\ell), (\mathbf{w}_1, \dots, \mathbf{w}_\ell)$) $\rightarrow \pi$:
1. **Interpolate wire values.** Let $v_{i,j}$ denote the value of j -th wire in $C(\mathbf{x}_i, \mathbf{w}_i)$ and $\mathbf{v}_j = (v_{1,j}, \dots, v_{\ell,j})$ denote the vector of all values for the j -th wire (across all ℓ instances). For each $j \in [t]$, let $\phi_j(X)$ denote the unique polynomial of degree at most $\ell - 1$ which evaluates to $\mathbf{v}_j$ over $\mathbb{L}$, i.e., $\phi_j(\omega_L^i) = v_{i,j}$ for all $i \in [\ell]$. The wire value matrix $\mathbf{V}$ is a $\ell \times t$ matrix where the j -th column is $\mathbf{v}_j$, i.e., $\mathbf{V}[i, j] = \mathbf{v}_j[i] = v_{i,j}$. Let $\phi(X, Y) = \sum_{j \in [t]} \phi_j(X) \bar{L}_j(Y)$ be the unique bivariate polynomials of degrees $\deg_X < \ell$ and $\deg_Y < t$ that evaluates to $\mathbf{V}$ over $\mathbb{L} \times \mathbb{V}$. For each NAND gate $G_k = (\mathbf{a}_k, \mathbf{b}_k, \mathbf{c}_k)$ where $k \in [s]$, let $f_k^{(1)}(X), f_k^{(2)}(X), f_k^{(3)}(X)$ be polynomials of degree at most $\ell - 1$ which evaluate to $\mathbf{a}_k, \mathbf{b}_k, \mathbf{c}_k$ over $\mathbb{L}$ respectively, where the values $f_k^{(1)}(\omega_L^i), f_k^{(2)}(\omega_L^i)$, and $f_k^{(3)}(\omega_L^i)$ represent the values of the left input, right input, and output wires of the k -th gate in $C(\mathbf{x}_i, \mathbf{w}_i)$, respectively. The wire value matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are three $\ell \times s$ matrices where the k -th column of each corresponds to the vectors $\mathbf{a}_k, \mathbf{b}_k, \mathbf{c}_k$ respectively, i.e., $\mathbf{A}[k, i] = \mathbf{a}_k[i], \mathbf{B}[k, i] = \mathbf{b}_k[i], \mathbf{C}[k, i] = \mathbf{c}_k[i]$. Similarly, let $f^{(1)}(X, Z), f^{(2)}(X, Z), f^{(3)}(X, Z)$ be the unique polynomial of degrees $\deg_X < \ell$ and $\deg_Z < s$ that evaluates to $\mathbf{A}, \mathbf{B}, \mathbf{C}$ over $\mathbb{L} \times \mathbb{H}$, respectively.
 2. **Encoding these polynomials.** Compute each $\phi(X, Y), \{f^{(j)}(X, Z)\}_{j \in [3]}$ using Lagrange interpolation. Set $\phi = \prod_{i \in [t], j \in [\ell]} [\phi(\omega_L^i, \omega_V^j) L_i(\tau_X) \bar{L}_j(\tau_Y)]_1$ as the commitment to $\mathbf{V}$. Similarly, for $j \in [3]$, the commitment to the matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ is set as $F^{(j)} = \prod_{i \in [t], k \in [s]} [f^{(j)}(\omega_L^i, \omega_H^k) L_i(\tau_X) \tilde{L}_k(\tau_Z)]_1$.
 3. **Validity of wire assignments.** Let $\rho(R, Y) = \sum_{j \in [t]} \rho_j(R) \bar{L}_j(Y)$ be a prescribed polynomial such that all $\rho(R, \omega_V^j)$ for $\omega_V^j \in \mathbb{V}$ are $|\mathbb{V}|$ linearly independent polynomials and let $\Phi(X, Y) = \phi(X, Y) \phi(X, Y) - \phi(X, Y)$. Denote $\Phi^*(X, R) = \sum_{\omega_V^j \in \mathbb{V}} \Phi(X, \omega_V^j) \rho(R, \omega_V^j)$. The relation is enforced in the following steps:

(a) $\forall \omega_L^i \in \mathbb{L}, \Phi^*(\omega_L^i, R) = 0$, i.e., there exists $\eta(X)$ satisfying

$$\Phi^*(X, R) = \eta(X) \cdot \zeta_{\mathbb{L}}(X).$$

Set $\Phi^* = [\Phi^*(\tau_X, \tau_R)]_1, \eta = [\eta(\tau_X)]_1$.

(b) $\Phi^*(X, R) = \sum_{\omega_L^j \in \mathbb{V}} \Phi(X, \omega_L^j) \cdot \rho(R, \omega_L^j)$. Run

$$u \leftarrow \text{BuLosum.Prove}(\text{crs}, \Phi(X, Y), \mathbb{V}, \mathbb{L})$$

which guarantees the existence of u satisfying

$$\Phi(\tau_X, \tau_Y) \cdot \rho(\tau_R, \tau_Y) - \Phi^*(\tau_X, \tau_R) \bar{L}_{j^*}(\tau_Y) = u(\tau_X, \tau_Y) \cdot \gamma^{-1}$$

Set $\Phi = [\Phi(\tau_X, \tau_Y)]_1, \tilde{\phi} = [\phi(\tau_X, \tau_Y)]_2$. Set $\pi_1 = (\Phi^*, \eta, \tilde{\phi}, u, \Phi)$.

4. **Validity of gate computation.** Let $\bar{\rho}(R, Z) = \sum_{k \in [s]} \bar{\rho}_k(R) \tilde{L}_k(Z)$ be a prescribed polynomial such that all $\bar{\rho}(R, \omega_H^k)$ for $\omega_H^k \in \mathbb{H}$ are $|\mathbb{H}|$ linearly independent polynomials and let $G(X, Z) = 1 - f^{(1)}(X, Z) f^{(2)}(X, Z) - f^{(3)}(X, Z)$. Denote $G^*(X, R) = \sum_{\omega_H^k \in \mathbb{H}} G(X, \omega_H^k) \bar{\rho}(R, \omega_H^k)$. The relation is equivalent to the following two steps:

(a) $\forall \omega_L^i \in \mathbb{L}, G^*(\omega_L^i, R) = 0$, i.e., there exists $\bar{\eta}(X)$ satisfying

$$G^*(X, R) = \bar{\eta}(X) \cdot \zeta_{\mathbb{L}}(X).$$

Set $G^* = [G^*(\tau_X, \tau_R)]_1$ and $\bar{\eta} = [\bar{\eta}(\tau_X)]_1$.

(b) $G^*(X, R) = \sum_{\omega_H^k \in \mathbb{H}} G(X, \omega_H^k) \cdot \bar{\rho}(R, \omega_H^k)$. Run

$$\bar{u} \leftarrow \text{BuLosum.Prove}(\text{crs}, G(X, Z), \mathbb{H}, \mathbb{L})$$

which guarantees the existence of $\bar{u}$ satisfying

$$G(\tau_X, \tau_Z) \cdot \bar{\rho}(\tau_R, \tau_Z) - G^*(\tau_X, \tau_R) \tilde{L}_{k^*}(\tau_Z) = \bar{u}(\tau_X, \tau_Z) \cdot \bar{\gamma}^{-1}.$$

Set $G = [G(\tau_X, \tau_Z)]_1, \tilde{F} = [f^{(2)}(\tau_X, \tau_Z)]_2$. Set $\pi_2 = (G^*, \bar{\eta}, G, \tilde{F}, \bar{u})$.

5. **Validity of gate permutation.** Let $I_1 \subset [t]$ be the set of indices in $(\mathbf{v}_1, \dots, \mathbf{v}_t)$ that $(\mathbf{a}_1, \dots, \mathbf{a}_s)$ takes from and $\mu_1 : [s] \rightarrow I_1$ be a mapping such that $\mathbf{a}_k = \mathbf{v}_{\mu_1(k)}$ for all $k \in [s]$. Similarly, let $I_2 \subset [t]$ (resp. $I_3 \subset [t]$) be the set of indices in $(\mathbf{v}_1, \dots, \mathbf{v}_t)$ that $(\mathbf{b}_1, \dots, \mathbf{b}_s)$ (resp. $(\mathbf{c}_1, \dots, \mathbf{c}_s)$) takes from and $\mu_2 : [s] \rightarrow I_2$ (resp. $\mu_3 : [s] \rightarrow I_3$) be a mapping such that $\mathbf{b}_k = \mathbf{v}_{\mu_2(k)}$ (resp. $\mathbf{c}_k = \mathbf{v}_{\mu_3(k)}$) for all $k \in [s]$. Then, for each NAND gate $G_k = (\mathbf{a}_k, \mathbf{b}_k, \mathbf{c}_k)$ of circuit C where $k \in [s]$, there exist selection matrices $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$ of size $t \times s$ that select each NAND gate's left input, right input and output. Specifically,

- $\mathbf{P}_1$ is a public binary matrix where for each $k \in [s]$, $\mathbf{P}_1[\mu_1(k), k] = 1$ and all other entries are 0. Thus, $\mathbf{A} = \mathbf{V} \cdot \mathbf{P}_1$.

- $\mathbf{P}_2$ is a public binary matrix where for each $k \in [s]$, $\mathbf{P}_2[\mu_2(k), k] = 1$ and all other entries are 0. Thus, $\mathbf{B} = \mathbf{V} \cdot \mathbf{P}_2$.
- $\mathbf{P}_3$ is a public binary matrix where for each $k \in [s]$, $\mathbf{P}_3[\mu_3(k), k] = 1$ and all other entries are 0. Thus, $\mathbf{C} = \mathbf{V} \cdot \mathbf{P}_3$.

Let $P^{(1)}(Y, Z), P^{(2)}(Y, Z), P^{(3)}(Y, Z)$ be polynomials with $\deg_Y < t$ and $\deg_Z < s$, such that their evaluations over $\mathbb{V} \times \mathbb{H}$ equal the matrices $\mathbf{P}_1[j, k], \mathbf{P}_2[j, k], \mathbf{P}_3[j, k]$, respectively. Particularly, each $\mathbf{P}_i$ is a sparse matrix since they only have s nonzero entries. To check these three linear constraints, run MatrixMul+ to generate

$$\begin{aligned}\hat{\pi}_1 &= (\hat{\pi}_{1,1}, \hat{\pi}_{1,2}, \hat{\pi}_{1,3}) \leftarrow \text{MatrixMul+}.\text{Prove}(\text{crs}, \mathbf{P}_1, \mathbf{A}, \mathbf{V}), \\ \hat{\pi}_2 &= (\hat{\pi}_{2,1}, \hat{\pi}_{2,2}, \hat{\pi}_{2,3}) \leftarrow \text{MatrixMul+}.\text{Prove}(\text{crs}, \mathbf{P}_2, \mathbf{B}, \mathbf{V}), \\ \hat{\pi}_3 &= (\hat{\pi}_{3,1}, \hat{\pi}_{3,2}, \hat{\pi}_{3,3}) \leftarrow \text{MatrixMul+}.\text{Prove}(\text{crs}, \mathbf{P}_3, \mathbf{C}, \mathbf{V}).\end{aligned}$$

Compute $\theta = H(\text{crs}, \phi, \pi_1, \pi_2, F^{(1)}, F^{(2)}, F^{(3)})$ and set $\pi_3 = \hat{\pi}_1 + \theta \hat{\pi}_2 + \theta^2 \hat{\pi}_3$. We can merge these three matrix multiplication checks into a single matrix multiplication check.

6. **Output satisfiability.** We prove that the output is a commitment to the all-ones vector by showing $\phi(X, \omega_V^t) = \phi_t(X) \equiv 1$. Equivalently, there exists a polynomial $q_t(X, Y)$ such that

$$\phi(X, Y) - 1 = q_t(X, Y) (Y - t).$$

Set $\pi_4 = q_t = [q_t(\tau_X, \tau_Y)]_1$.

7. Output proof $\pi = (\phi, \pi_1, F^{(1)}, F^{(2)}, F^{(3)}, \pi_2, \pi_3, \pi_4)$
- $\text{Verify}(\text{crs}, C, (\mathbf{x}_1, \dots, \mathbf{x}_\ell), \pi) \rightarrow \{0, 1\}$: This algorithm checks the following properties:
- **Validity of wire assignments.** Check that

$$\begin{aligned}e(\Phi^*, [1]_2) &= e([\eta]_1, [\zeta_{\mathbb{L}}(\tau_X)]_2), \quad e(\phi, [1]_2) = e([1]_1, \tilde{\phi}), \\ e(\Phi, [1]_2) &= e(\phi, \tilde{\phi}) \cdot e(-\phi, [1]_2)\end{aligned}$$

and run

$$1 \leftarrow \text{BuLosum}.\text{Verify}(\text{crs}, \Phi, \Phi^*, u)$$

which with CRS elements $[\bar{L}_{j^*}(\tau_Y)]_2, [\gamma^{-1}]_2$ implies

$$e(\Phi, [\rho(\tau_R, \tau_Y)]_2) \cdot e(-\Phi^*, [\bar{L}_{j^*}(\tau_Y)]_2) = e(u, [\gamma^{-1}]_2).$$

- **Validity of gate computation.** Check that

$$\begin{aligned}e(G^*, [1]_2) &= e([\bar{\eta}]_1, [\zeta_{\mathbb{L}}(\tau_X)]_2), \quad e(F^{(2)}, [1]_2) = e([1]_1, \tilde{F}), \\ e(G, [1]_2) \cdot e(F^{(1)}, \tilde{F}) \cdot e(F^{(3)}, [1]_2) &= e([1]_1, [1]_2)\end{aligned}$$

and run

$$1 \leftarrow \text{BuLosum.Verify}(\text{crs}, G, G^*, \bar{u})$$

which, with CRS elements $[\tilde{L}_{k^*}(\tau_Z)]_2$ and $[\bar{y}^{-1}]_2$, implies

$$e(G, [\bar{\rho}(\tau_R, \tau_Z)]_2) \cdot e(-G^*, [\tilde{L}_{k^*}(\tau_Z)]_2) = e(\bar{u}, [\bar{y}^{-1}]_2).$$

- **Validity of gate permutation.** Compute $\theta = H(\text{crs}, \phi, \pi_1, \pi_2, F^{(1)}, F^{(2)}, F^{(3)})$, $P = P_1 + \theta P_2 + \theta^2 P_3$ and $F = F^{(1)} + \theta F^{(2)} + \theta F^{(3)}$. Run

$$1 \leftarrow \text{MatrixMul+.Verify}(\text{crs}, \pi_3, P, F, \phi).$$

- **Output satisfiability.** Check that

$$e(\phi, [1]_2) = e([1]_1, [1]_2) \cdot e(\pi_4, [(\tau_Y - t)]_2).$$

- Output 1 if all checks pass and 0 otherwise.

5.2 Completeness.

Theorem 3 (Completeness). *For any Boolean circuit C and any set of ℓ instance-witness pairs $\{(\mathbf{x}_i, \mathbf{w}_i)\}_{i=1}^\ell$ such that $C(\mathbf{x}_i, \mathbf{w}_i) = 1$ for all $i \in [\ell]$, if the prover follows the introduced protocol BARG correctly, then:*

$$\Pr [\text{Verify}(\text{crs}, C, (\mathbf{x}_1, \dots, \mathbf{x}_\ell), \text{Prove}(\text{crs}, C, (\mathbf{x}_1, \dots, \mathbf{x}_\ell), (\mathbf{w}_1, \dots, \mathbf{w}_\ell))) = 1] = 1.$$

Proof. We prove completeness by demonstrating that each verification check passes when the prover honestly executes the protocol with valid inputs.

1. *Interpolation and commitments.* The honest prover constructs

$$\phi(X, Y) = \sum_{i \in [t]} \phi_i(X) \bar{L}_i(Y),$$

where each $\phi_i(X)$ is the unique polynomial of degree $< \ell$ interpolating the i -th wire values on $\mathbb{L}$; similarly $f^{(j)}(X, Z)$ interpolates $\mathbf{A}, \mathbf{B}, \mathbf{C}$, respectively. The prover computes KZG-style commitments

$$\begin{aligned} \phi &= [\phi(\tau_X, \tau_Y)]_1, F^{(j)} = [f^{(j)}(\tau_X, \tau_Z)]_1, \Phi^* = [\Phi^*(\tau_X, \tau_R)]_1, \tilde{\phi} = [\tilde{\phi}(\tau_X, \tau_Y)]_2, \\ \tilde{F} &= [f^{(2)}(\tau_X, \tau_Z)]_2, G = [G(\tau_X, \tau_Z)]_1, G^* = [G^*(\tau_X, \tau_R)]_1, q_t = [q_t(\tau_X, \tau_Y)]_1. \end{aligned}$$

Since commitments are formed honestly from the polynomials, the commitment and opening relations used in Verify algorithm hold exactly by the completeness property of the underlying polynomial commitment scheme.

2. Validity of wire assignments. The verification checks for wire assignments are: Define

$$\Phi(X, Y) = \phi(X, Y)\phi(X, Y) - \phi(X, Y).$$

For an honest assignment where each wire value is 0 or 1, we have $\Phi(\omega_L^i, \omega_V^j) = 0$ for every $\omega_L^i \in \mathbb{L}, \omega_V^j \in \mathbb{V}$. Define

$$\Phi^*(X, R) = \sum_{\omega_V^j \in \mathbb{V}} \Phi(X, \omega_V^j) \rho(R, \omega_V^j),$$

and produces $\eta(X)$ with $\Phi^*(X, R) = \eta(X) \zeta_{\mathbb{L}}(X)$. Since the prover computed Φ and Φ^* from the true polynomials, both the algebraic identity and the divisibility relation hold. By the completeness of BuLosum, the produced sub-proof

$$u \leftarrow \text{BuLosum.Prove}(\text{crs}, \Phi(X, Y), \mathbb{V}, \mathbb{L})$$

satisfies $\text{BuLosum.Verify}(\text{crs}, \Phi, \Phi^*, u) = 1$. Thus the wire-assignment checks

$$e(\Phi^*, [1]_2) = e([\eta]_1, [\zeta_{\mathbb{L}}(\tau_X)]_2), \quad \text{BuLosum.Verify}(\text{crs}, \Phi, \Phi^*, u) = 1,$$

and the pairing consistency checks involving $\phi, \Phi, \tilde{\phi}$ all hold.

3. Validity of gate computation. The verification checks for gate computation are: For every gate, define

$$G(X, Z) = 1 - f^{(1)}(X, Z)f^{(2)}(X, Z) - f^{(3)}(X, Z),$$

and forms

$$G^*(X, R) = \sum_{\omega_H^k \in \mathbb{H}} G(X, \omega_H^k) \bar{\rho}(R, \omega_H^k),$$

with $G^*(X, R) = \bar{\eta}(X) \zeta_{\mathbb{L}}(X)$. For honest gate values (i.e. each gate satisfies the NAND relation on the instances), we have $G(\omega_L^i, \omega_H^k) \equiv 0$ on $\mathbb{L} \times \mathbb{H}$, hence the divisibility and the BuLosum identity hold. By the completeness of BuLosum, the prover's subproof $\bar{u} \leftarrow \text{BuLosum.Prove}(\text{crs}, G(X, Z), \mathbb{H}, \mathbb{L})$ verifies, so the verifier's gate-computation checks pass:

$$e(G^*, [1]_2) = e([\bar{\eta}]_1, [\zeta_{\mathbb{L}}(\tau_X)]_2), \quad \text{BuLosum.Verify}(\text{crs}, G, G^*, \bar{u}) = 1,$$

and the pairings consistency checks involving $G, \{F^{(j)}\}_{j \in [3]}$ and $\tilde{F}$ all hold.

4. Validity of gate permutation. By construction of selection matrices $\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3$ and the definitions $\mathbf{A} = \mathbf{V}\mathbf{P}_1, \mathbf{B} = \mathbf{V}\mathbf{P}_2, \mathbf{C} = \mathbf{V}\mathbf{P}_3$, the honest prover supplies the correct matrices and honest MatrixMul+ subproofs:

$$\hat{\pi}_i \leftarrow \text{MatrixMul+.Prove}(\text{crs}, \mathbf{P}_i, \mathbf{X}_i, \mathbf{V}), \quad i = 1, 2, 3,$$

where $\mathbf{X}_1 = \mathbf{A}$, $\mathbf{X}_2 = \mathbf{B}$, $\mathbf{X}_3 = \mathbf{C}$. Because MatrixMul+ is complete, each of these sub-proofs verifies. After the linear combination using $\theta = H(\cdot)$, the combined proof π_3 still corresponds to valid matrix-multiplication relations, and therefore

$$\text{MatrixMul+}.Verify(\text{crs}, \pi_3, P, F, \phi) = 1$$

holds for the verifier.

5. Output satisfiability. The prover sets the polynomials such that for the output wire index (the designated ω_V^t), $\phi(\omega_L^i, \omega_V^t) \equiv 1$ for each $\omega_L^i \in \mathbb{L}$, so there exists $q_t(X, Y)$ satisfying

$$\phi(X, Y) - 1 = q_t(X, Y) (Y - t).$$

The prover computes the commitment $\pi_4 = [q_t(\tau_X, \tau_Y)]_1$ honestly. Using the KZG completeness property and bilinearity, the pairing identity

$$e(\phi, [1]_2) = e([1]_1, [1]_2) \cdot e(\pi_4, [(\tau_Y - t)]_2)$$

holds for honest openings, and the verifier's output check passes.

All four categories of checks (wire validity, gate computation, permutation, and output-satisfiability) are satisfied when the prover is honest and the underlying protocols are complete. Therefore the verifier accepts with probability 1. This completes the proof of completeness. $\square$

5.3 Security

Theorem 4 (Soundness). *Assume that:*

1. *The underlying polynomial commitment scheme KZG is knowledge sound with advantage $\epsilon_{\text{SOUND}}^{\text{KZG}}(\lambda)$.*
2. *The BuLosum protocol is sound with advantage $\epsilon_{\text{SOUND}}^{\text{BuLosum}}(\lambda)$.*
3. *The MatrixMul+ protocol is sound with advantage $\epsilon_{\text{SOUND}}^{\text{MatrixMul+}}(\lambda)$.*

Then for every PPT adversary $\mathcal{A}_{\text{BARG}}$ and instances $\mathbf{x} = (C, \mathbf{x}_{1:\ell})$ where $C : \{0, 1\}^n \times \{0, 1\}^h \rightarrow \{0, 1\}$ of size at most $s(\lambda)$, there exist PPT adversaries $\mathcal{B}_{\text{BuLosum}}$, $\mathcal{B}_{\text{MatrixMul+}}$, and $\mathcal{B}_{\text{KZG}}$ such that

$$\epsilon_{\text{SOUND}}^{\text{BARG}}(\mathcal{A}_{\text{BARG}}) =$$

$$\Pr \left[(\pi \leftarrow \mathcal{A}_{\text{BARG}}(\text{crs}, \mathbf{x}_{1:\ell})) : \text{Verify}(\text{crs}, \mathbf{x}_{1:\ell}, \pi) = 1 \wedge \exists j \in [\ell], C(\mathbf{x}_j, \mathbf{w}_j) \neq 1 \right] \leq$$

$$\epsilon_{\text{SOUND}}^{\text{BuLosum}}(\mathcal{B}_{\text{BuLosum}}) + \epsilon_{\text{SOUND}}^{\text{MatrixMul+}}(\mathcal{B}_{\text{MatrixMul+}}) + \epsilon_{\text{SOUND}}^{\text{KZG}}(\mathcal{B}_{\text{KZG}}).$$

Hence, if all three underlying advantages are negligible in λ , the introduced protocol BARG is sound.

Proof. We prove soundness through a sequence of games, showing that any adversary breaking the soundness of our scheme can be used to break the security of one of the underlying components.

$\mathbf{G}_0$ (*real game*). The challenger samples crs honestly for all sub-protocols and gives it to $\mathcal{A}_{\text{BARG}}$. Adversary $\mathcal{A}_{\text{BARG}}$ outputs a purported proof π . The verifier $\text{Verify}(\text{crs}, C, (\mathbf{x}_1, \dots, \mathbf{x}_\ell), \pi)$ runs all checks (*wire, gate, permutation, output*) exactly as specified in BARG.

1. The challenger runs $\text{Setup}(\lambda, \ell, t, s) \rightarrow \text{crs}$ and gives crs to adversary $\mathcal{A}_{\text{BARG}}$.
2. $\mathcal{A}_{\text{BARG}}$ outputs a circuit C , instances $(\mathbf{x}_1, \dots, \mathbf{x}_\ell)$, and a proof π
3. $\mathcal{A}_{\text{BARG}}$ wins if $\text{Verify}(\text{crs}, C, (\mathbf{x}_1, \dots, \mathbf{x}_\ell), \pi) = 1$ and $\exists j \in [\ell]$ such that $C(\mathbf{x}_j, \mathbf{w}_j)$ is unsatisfiable.

Let $\text{Acc}(\mathbf{G}_i)$ denote the probability that $\mathcal{A}_{\text{BARG}}$ wins in G_i . Thus, $\text{Acc}(\mathbf{G}_0) = \epsilon_{\text{SOUND}}^{\text{BARG}}(\mathcal{A}_{\text{BARG}})$.

$\mathbf{G}_1$ (*BuLosum verification*). We replace the real BuLosum verifier with an ideal oracle that rejects whenever the underlying polynomial identity tested by BuLosum is false. $\mathcal{B}_{\text{BuLosum}}$ internally runs $\mathcal{A}_{\text{BARG}}$ and forwards any BuLosum sub-proofs π_1^* and π_2^* to its own BuLosum challenger. If $\mathcal{A}_{\text{BARG}}$ produces a proof that makes the real verifier accept but some gate computation or wire assignment is incorrect, then $\mathcal{B}_{\text{BuLosum}}$ wins the BuLosum soundness game. Therefore,

$$|\text{Acc}(\mathbf{G}_0) - \text{Acc}(\mathbf{G}_1)| \leq \epsilon_{\text{SOUND}}^{\text{BuLosum}}(\mathcal{B}_{\text{BuLosum}}).$$

$\mathbf{G}_2$ (*MatrixMul+ verification*). Next, we replace the real MatrixMul+ verification with an ideal oracle enforcing the corresponding matrix identity. $\mathcal{B}_{\text{MatrixMul+}}$ runs $\mathcal{A}_{\text{BARG}}$, and when $\mathcal{A}_{\text{BARG}}$ outputs sub-proofs π_3^* for MatrixMul+, $\mathcal{B}_{\text{MatrixMul+}}$ forwards them to its own challenger. If $\mathcal{A}_{\text{BARG}}$ convinces the real verifier while the wire permutation is incorrect, $\mathcal{B}_{\text{MatrixMul+}}$ breaks MatrixMul+ soundness. Hence,

$$|\text{Acc}(\mathbf{G}_1) - \text{Acc}(\mathbf{G}_2)| \leq \epsilon_{\text{SOUND}}^{\text{MatrixMul+}}(\mathcal{B}_{\text{MatrixMul+}}).$$

$\mathbf{G}_3$ (*KZG soundness*). At this point all BuLosum and MatrixMul+ subchecks that pass correspond to true polynomial or matrix equalities on the prescribed domains. The only remaining way for $\mathcal{A}_{\text{BARG}}$ to make Verify accept a false instance is to produce inconsistent commitments that still satisfy all pairing checks, thus violating the knowledge soundness property of KZG. The challenger can extract the polynomials underlying all commitments:

- $\phi(X, Y)$ from commitment ϕ ,
- $f^{(1)}(X, Z), f^{(2)}(X, Z), f^{(3)}(X, Z)$ from $F^{(1)}, F^{(2)}, F^{(3)}$
- $\eta(X), \bar{\eta}(X), \Phi(X, Y), \Phi^*(X, Y), G(X, Z), G^*(X, Z), q_t(X, Y)$ from their respective commitments

Particularly, the final output check also relies on the KZG polynomial commitment scheme, where the sub-proof constitutes a KZG opening proof at $Y = t$ to prove the relation

$$\phi(X, Y) - 1 = q_t(X, Y) \cdot (Y - t).$$

Thus, the security of KZG guarantees this polynomial identity holds identically.

$\mathcal{B}_{\text{KZG}}$ runs $\mathcal{A}_{\text{BARG}}$ and extracts from the accepting transcript two distinct openings for the same commitment. Therefore

$$|\text{Acc}(\mathbf{G}_2) - \text{Acc}(\mathbf{G}_3)| \leq \epsilon_{\text{SOUND}}^{\text{KZG}}(\mathcal{B}_{\text{KZG}}).$$

In $\mathbf{G}_3$, all commitments satisfy knowledge soundness, so any accepting transcript implies that all polynomial constraints (wire, gate, permutation, and output) hold simultaneously. When $\mathbf{x}_j$ is unsatisfiable, the probability of $\mathcal{A}_{\text{BARG}}$ winning in $\mathbf{G}_3$ is exactly zero, i.e., $\text{Acc}(\mathbf{G}_3) = 0$.

Final bound. Combining all game transitions:

$$\begin{aligned} \text{Acc}(\mathbf{G}_0) &\leq \text{Acc}(\mathbf{G}_3) + \epsilon_{\text{SOUND}}^{\text{BuLosum}}(\mathcal{B}_{\text{BuLosum}}) + \epsilon_{\text{SOUND}}^{\text{MatrixMul+}}(\mathcal{B}_{\text{MatrixMul+}}) + \epsilon_{\text{SOUND}}^{\text{KZG}}(\mathcal{B}_{\text{KZG}}) \\ &= \epsilon_{\text{SOUND}}^{\text{BuLosum}}(\mathcal{B}_{\text{BuLosum}}) + \epsilon_{\text{SOUND}}^{\text{MatrixMul+}}(\mathcal{B}_{\text{MatrixMul+}}) + \epsilon_{\text{SOUND}}^{\text{KZG}}(\mathcal{B}_{\text{KZG}}) \end{aligned}$$

which yields the claimed bound. $\square$

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Appendix

A Committed Matrix Multiplication

In this section, we use the variant of bi-to-uni sumcheck lemma 3 to construct an argument system MatrixMul for proving that a matrix $\mathbf{U} \in \mathbb{Z}_p^{\ell \times s}$ equals the product of $\mathbf{P} \in \mathbb{Z}_p^{t \times s}$ and $\mathbf{V} \in \mathbb{Z}_p^{\ell \times t}$.

A.1 Construction

The formal description of MatrixMul is given below.

– Setup(λ, ℓ, t, s) $\rightarrow$ crs:

1. Run $\mathcal{G} = (p, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T, e) \leftarrow \text{GroupGen}(\lambda)$.
2. Denote three multiplicative subgroups $\mathbb{L} = \{\omega_L, \dots, \omega_L^\ell\}$, $\mathbb{V} = \{\omega_V, \dots, \omega_V^t\}$, and $\mathbb{H} = \{\omega_H, \dots, \omega_H^s\}$ such that $|\mathbb{L}| = \ell$, $|\mathbb{V}| = t$, $|\mathbb{H}| = s$.
3. Define $L_i(X) = \frac{\omega_L^i}{|\mathbb{L}|} \cdot \frac{X^{|\mathbb{L}|-1}}{X - \omega_L^i}$, $\bar{L}_j(Y) = \frac{\omega_V^j}{|\mathbb{V}|} \cdot \frac{Y^{|\mathbb{V}|-1}}{Y - \omega_V^j}$ and $\tilde{L}_k(Z) = \frac{\omega_H^k}{|\mathbb{H}|} \cdot \frac{Z^{|\mathbb{H}|-1}}{Z - \omega_H^k}$.
4. Sample $\tau_X, \tau_Y, \tau_Z \leftarrow \mathbb{Z}_p$. Sample a random vector $\mathbf{r} = (r_1, \dots, r_\ell) \in \mathbb{Z}_p^\ell$. Output crs as

$$\left(\mathcal{G}, [\tau_Y]_2, [\tau_Z]_2, \left\{ [\tau_X^i \tau_Y^j]_1, [\tau_X^i \tau_Z^k]_1, [\tau_Y^j \tau_Z^k]_1 \right\}_{i \in [0, \ell-1], j \in [0, t-1], k \in [0, s-1]}, \right. \\ \left. \left\{ r_i L_i(\tau_X) \right\}_2, [r_i \bar{L}_j(\tau_Y)]_2, [r_i \tilde{L}_k(\tau_Z)]_2, \left[r_i \bar{L}_j(\tau_Y) \frac{L_i(\tau_X) - L_i(0)}{\tau_X} \right]_1, \right. \\ \left[r_i \tilde{L}_k(\tau_Z) \frac{L_i(\tau_X) - L_i(0)}{\tau_X} \right]_1, \left[r_i \tilde{L}_k(\tau_Z) \frac{\bar{L}_j(\tau_Y) - \bar{L}_j(0)}{\tau_Y} \right]_1, \\ \left[r_i \tilde{L}_k(Z) \frac{\bar{L}_j(Y) \bar{L}_{j'}(Y)}{\zeta_V(Y)} \right]_1, \left[r_{i'} \bar{L}_j(Y) \frac{L_i(X) L_{i'}(X)}{\zeta_L(X)} \right]_1, \\ \left. \left[r_{i'} \tilde{L}_k(Z) \frac{L_i(X) L_{i'}(X)}{\zeta_L(X)} \right]_1 \right\}_{i, i' \in [\ell], j, j' \in [t], k \in [s], j \neq j', i \neq i'} , [r(\tau_X)]_2. \right)$$

– Prove(crs, $\mathbf{P}, \mathbf{U}, \mathbf{V}$) : Takes as input crs and three matrices $\mathbf{P} \in \mathbb{Z}_p^{t \times s}$, $\mathbf{U} \in \mathbb{Z}_p^{\ell \times s}$, $\mathbf{V} \in \mathbb{Z}_p^{\ell \times t}$, then proceed as follows:

1. Compute three polynomials $P(Y, Z)$, $V(X, Y)$ and $U(X, Z)$ by interpolating $\mathbf{P}[j, k]$ on all (ω_V^j, ω_H^k) , $\mathbf{V}[i, j]$ on all (ω_L^i, ω_V^j) and $\mathbf{U}[i, k]$ on all (ω_L^i, ω_H^k) , respectively. Set

$$P = \left[\sum_{k=1}^s \sum_{j=1}^t \mathbf{P}[j, k] \bar{L}_j(\tau_Y) \tilde{L}_k(\tau_Z) \right]_1, V = \left[\sum_{j=1}^t \sum_{i=1}^\ell \mathbf{V}[i, j] L_i(\tau_X) \bar{L}_j(\tau_Y) \right]_1, \\ U = \left[\sum_{k=1}^s \sum_{i=1}^\ell \mathbf{U}[i, k] L_i(\tau_X) \tilde{L}_k(\tau_Z) \right]_1.$$

2. Set the intermediate vectors $\mathbf{d}, \mathbf{h}$ as

$$[d(\tau_Z)]_2 = \sum_{i=1}^{\ell} \sum_{k=1}^s U(\omega_L^i, \omega_H^k) [r_i \tilde{L}_k(\tau_Z)]_2,$$

$$[h(\tau_Y)]_2 = \sum_{i=1}^{\ell} \sum_{j=1}^t V(\omega_L^i, \omega_V^j) [r_i \bar{L}_j(\tau_Y)]_2.$$

3. Compute polynomials $Q(X, Y)$, $\bar{Q}(X, Z)$ and $\tilde{Q}(Y, Z)$ such that

$$V(X, Y) \cdot r(X) = \frac{h(Y)}{\ell} + Q(X, Y) \cdot X + D(X, Y) \cdot \zeta_{\mathbb{L}}(X),$$

$$U(X, Z) \cdot r(X) = \frac{d(Z)}{\ell} + \bar{Q}(X, Z) \cdot X + \bar{D}(X, Z) \cdot \zeta_{\mathbb{L}}(X),$$

$$P(Y, Z) \cdot h(Y) = \frac{d(Z)}{t} + \tilde{Q}(Y, Z) \cdot Y + \tilde{D}(Y, Z) \cdot \zeta_{\mathbb{V}}(Y)$$

where $r(X) = \sum_{i=1}^{\ell} r_i L_i(X)$. According to Lemma 3, these polynomials can be computed as:

$$Q(X, Y) = \sum_{j=1}^t \sum_{i=1}^{\ell} V(\omega_L^i, \omega_V^j) \left(r_i \bar{L}_j(Y) \frac{L_i(X) - L_i(0)}{X} \right),$$

$$\bar{Q}(X, Z) = \sum_{k=1}^s \sum_{i=1}^{\ell} U(\omega_L^i, \omega_H^k) \left(r_i \tilde{L}_k(Z) \frac{L_i(X) - L_i(0)}{X} \right),$$

$$\tilde{Q}(Y, Z) = \sum_{k=1}^s \sum_{j=1}^t P(\omega_V^j, \omega_H^k) \sum_{i=1}^{\ell} V(\omega_L^i, \omega_V^j) \left(r_i \tilde{L}_k(Z) \frac{\bar{L}_j(Y) - \bar{L}_j(0)}{Y} \right),$$

$$D(X, Y) = \sum_{j=1}^t \sum_{i=1}^{\ell} V(\omega_L^i, \omega_V^j) \left(r_i \bar{L}_j(Y) \frac{L_i^2(X) - L_i(X)}{\zeta_{\mathbb{L}}(X)} \right) +$$

$$\sum_{j=1}^t \sum_{i=1}^{\ell} V(\omega_L^i, \omega_V^j) \sum_{i'=1, i' \neq i}^{\ell} \left(r_{i'} \bar{L}_j(Y) \frac{L_i(X) L_{i'}(X)}{\zeta_{\mathbb{L}}(X)} \right),$$

$$\bar{D}(X, Z) = \sum_{k=1}^s \sum_{i=1}^{\ell} U(\omega_L^i, \omega_H^k) \left(r_i \tilde{L}_k(Z) \frac{L_i^2(X) - L_i(X)}{\zeta_{\mathbb{L}}(X)} \right) +$$

$$\sum_{k=1}^s \sum_{i=1}^{\ell} V(\omega_L^i, \omega_H^k) \sum_{i'=1, i' \neq i}^{\ell} \left(r_{i'} \tilde{L}_k(Z) \frac{L_i(X) L_{i'}(X)}{\zeta_{\mathbb{L}}(X)} \right),$$

$$\tilde{D}(Y, Z) = \sum_{k=1}^s \sum_{j=1}^t P(\omega_V^j, \omega_H^k) \sum_{i=1}^{\ell} V(\omega_L^i, \omega_V^j) \left(r_i \tilde{L}_k(Z) \frac{\bar{L}_j^2(Y) - \bar{L}_j(Y)}{\zeta_{\mathbb{V}}(Y)} \right)$$

$$\sum_{k=1}^s \sum_{j=1}^t P(\omega_V^j, \omega_H^k) \sum_{i=1}^{\ell} \sum_{j'=1, j' \neq j}^t V(\omega_L^i, \omega_V^{j'}) \left(r_i \tilde{L}_k(Z) \frac{\bar{L}_j(Y) \bar{L}_{j'}(Y)}{\zeta_{\mathbb{V}}(Y)} \right).$$

where $h(\omega_V^j) = \sum_{i=1}^{\ell} r_i V(\omega_L^i, \omega_V^j)$.

4. Compute

$$\begin{aligned} D_{XY} &= [D(\tau_X, \tau_Y)]_1, \bar{D}_{XZ} = [\bar{D}(\tau_X, \tau_Z)]_1, \tilde{D}_{YZ} = [\tilde{D}(\tau_Y, \tau_Z)]_1, \\ Q_{XY} &= [Q(\tau_X, \tau_Y)]_1, \bar{Q}_{XZ} = [\bar{Q}(\tau_X, \tau_Z)]_1, \tilde{Q}_{YZ} = [\tilde{Q}(\tau_Y, \tau_Z)]_1, \\ q_{XY} &= [Q(\tau_X, \tau_Y) \cdot \tau_X]_1, \bar{q}_{XZ} = [\bar{Q}(\tau_X, \tau_Z) \cdot \tau_X]_1, \tilde{q}_{YZ} = [\tilde{Q}(\tau_Y, \tau_Z) \cdot \tau_Y]_1. \end{aligned}$$

5. Output $\pi = (D_{XY}, \bar{D}_{XZ}, \tilde{D}_{YZ}, Q_{XY}, \bar{Q}_{XZ}, \tilde{Q}_{YZ}, q_{XY}, \bar{q}_{XZ}, \tilde{q}_{YZ}, [d(\tau_Z)]_2, [h(\tau_Y)]_2)$.

– Verify(crs, π, P, V, U) $\rightarrow \{0, 1\}$:

1. Parse $\pi = (D_{XY}, \bar{D}_{XZ}, \tilde{D}_{YZ}, Q_{XY}, \bar{Q}_{XZ}, \tilde{Q}_{YZ}, q_{XY}, \bar{q}_{XZ}, \tilde{q}_{YZ}, [d(\tau_Z)]_2, [h(\tau_Y)]_2)$.

2. Verify bilinear pairing relation and output 1 iff

$$\begin{aligned} e(V, [r(\tau_X)]_2) &= e([1]_1, [h(\tau_Y)]_2) \cdot e(D_{XY}, [\tau_X^\ell - 1]_2) \cdot e(Q_{XY}, [\tau_X]_2) \wedge \\ e(U, [r(\tau_X)]_2) &= e([1]_1, [d(\tau_Z)]_2) \cdot e(\bar{D}_{XZ}, [\tau_X^\ell - 1]_2) \cdot e(\bar{Q}_{XZ}, [\tau_X]_2) \wedge \\ e(P, [h(\tau_Y)]_2) &= e([1]_1, [d(\tau_Z)]_2) \cdot e(\tilde{D}_{YZ}, [\tau_Y^t - 1]_2) \cdot e(\tilde{Q}_{YZ}, [\tau_Y]_2) \wedge \\ e(Q_{XY}, [\tau_X]_2) &= e(q_{XY}, [1]_2) \wedge e(\bar{Q}_{XZ}, [\tau_X]_2) = e(\bar{q}_{XZ}, [1]_2) \wedge \\ &e(\tilde{Q}_{YZ}, [\tau_Y]_2) = e(\tilde{q}_{YZ}, [1]_2). \end{aligned}$$

A.2 Performance

We analyze the theoretical performance of our MatrixMul protocol in terms of group operations. The Prover performs $\tilde{O}((s+t)\ell)$ field operations to interpolate the bivariate polynomials, and requires $O(st\ell)$ exponentiations in $\mathbb{G}_1$ to compute the proof elements $Q_{XY}, \bar{Q}_{XZ}, \tilde{Q}_{YZ}$ and $D_{XY}, \bar{D}_{XZ}, \tilde{D}_{YZ}$. The dominant cost arises from the computations of $\tilde{Q}_{YZ}$ and $\tilde{D}_{YZ}$, which involve a three-dimensional summation over all entries of the input matrices. In contrast, the verifier's running time is independent of the protocol parameters and is dominated by numerous pairings, highlighting the succinctness of verification.

For communication complexity, measured in group elements:

- $|\text{crs}| = O(st\ell)|\mathbb{G}_1| + O((s+t)\ell)|\mathbb{G}_2|,$
- $|\pi| = 9|\mathbb{G}_1| + 2|\mathbb{G}_2|.$

Thus, our work achieves *constant-size proofs* and *constant verification time*.

A.3 Completeness

The protocol satisfies perfect completeness if for all honestly generated CRS and valid inputs, the verification equations accept with probability 1. Let $\text{crs} \leftarrow \text{Setup}(\lambda, \ell, t, s)$ be a correctly generated common reference string and $\mathbf{P}, \mathbf{U}, \mathbf{V}$ be matrices satisfying the required relations. For

a proof $\pi \leftarrow \text{Prove}(\text{crs}, \mathbf{P}, \mathbf{U}, \mathbf{V})$, the corresponding polynomials satisfy the following algebraic identities:

$$\begin{aligned} V(X, Y) \cdot r(X) &= h(Y) + D(X, Y) \cdot \zeta_{\mathbb{L}}(X) + Q(X, Y) \cdot X, \\ U(X, Z) \cdot r(X) &= d(Z) + \bar{D}(X, Z) \cdot \zeta_{\mathbb{L}}(X) + \bar{Q}(X, Z) \cdot X, \\ P(Y, Z) \cdot h(Y) &= d(Z) + \tilde{D}(Y, Z) \cdot \zeta_{\mathbb{V}}(Y) + \tilde{Q}(Y, Z) \cdot Y, \\ q(X, Y) &= Q(X, Y) \cdot X, \bar{q}(X, Z) = \bar{Q}(X, Z) \cdot X, \tilde{q}(Y, Z) = \tilde{Q}(Y, Z) \cdot Y. \end{aligned}$$

Evaluating at the CRS elements (τ_X, τ_Y, τ_Z) and applying the bilinear map yields

$$\begin{aligned} e(V, [r(\tau_X)]_2) &= e([h(\tau_Y)]_2, [1]_2) \cdot e(D_{XY}, [\tau_X^\ell - 1]_2) \cdot e(Q_{XY}, [\tau_X]_2), \\ e(U, [r(\tau_X)]_2) &= e([d(\tau_Z)]_2, [1]_2) \cdot e(\bar{D}_{XZ}, [\tau_X^\ell - 1]_2) \cdot e(\bar{Q}_{XZ}, [\tau_X]_2), \\ e(P, [h(\tau_Y)]_2) &= e([d(\tau_Z)]_2, [1]_2) \cdot e(\tilde{D}_{YZ}, [\tau_Y^\ell - 1]_2) \cdot e(\tilde{Q}_{YZ}, [\tau_Y]_2), \\ e(Q_{XY}, [\tau_X]_2) &= e(q_{XY}, [1]_2), e(\bar{Q}_{XZ}, [\tau_X]_2) = e(\bar{q}_{XZ}, [1]_2), e(\tilde{Q}_{YZ}, [\tau_Y]_2) = e(\tilde{q}_{YZ}, [1]_2). \end{aligned}$$

Thus, any honestly generated proof is always accepted by the verifier, establishing the scheme's completeness. $\square$

B Security

In this section, we prove the security of MatrixMul in the generic group model of prime order. To do so, we rely on the “master theorems” from [39,40,41]. We start by recalling the generic group model.

Definition 2 (Generic Bilinear Group Model). Let $p \in \mathbb{Z}$ be a positive integer and let $S_i \subseteq \{0, 1\}^*$ and $i \in \{1, 2, \mathbb{T}\}$ be a set of strings of cardinality at least p . An arbitrary group $\mathbb{G}_i$ of prime order p is generically modeled by the random label function $\text{List}_i : \mathbb{Z}_p \rightarrow S_i$. We will think of $\text{List}_i(x)$ with $i \in \{1, 2, \mathbb{T}\}$ as corresponding to the string representation of g_i^x where $g_i \in \mathbb{G}_i$ is a fixed generator of the group $\mathbb{G}_i$. In the generic group model, the parities have access to the following queries:

- $\text{Label}(i, x)$ with $i \in \{1, 2, \mathbb{T}\}$ and value $x \in \mathbb{Z}_p$: if x has no label yet, allocate a fresh random string as its label $\text{List}_i(x) \leftarrow S_i$ and return the label $\text{List}_i(x)$.
- $\text{GroupOp}(i, a_1, a_2, u_1, u_2)$ with $a_1, a_2 \in \mathbb{F}_p$ and labels u_1, u_2 : If there exists $x_1, x_2 \in \mathbb{Z}_p$ such that $\text{List}_i(x_1) = u_1$ and $\text{List}_i(x_2) = u_2$, then return $\text{List}_i(a_1 x_1 + a_2 x_2)$; otherwise, return $\perp$.
- $\text{PairingOp}(u_1, u_2)$ with labels u_1, u_2 : If there exist $x_1, x_2 \in \mathbb{Z}_p$ such that $\text{List}_1(x_1) = u_1$, $\text{List}_2(x_2) = u_2$, then return $\text{List}_{\mathbb{T}}(x_1 \cdot x_2)$ in the target group; otherwise, return $\perp$.

Remark 1 (Symbolic model[42]). In the security proofs under the GGM, the challenger replaces each secret x_i with an indeterminate T_i , resulting in what is known as the “symbolic model”.

In this model, the label function's domain is expanded from $\mathbb{Z}_p$ to the polynomial ring $\mathbb{Z}_p[*]$. Proving the security in this model reduces to showing that a target polynomial is independent of the polynomials whose labels are present in the adversary's view. We will use this model to prove the security of MatrixMul.

Lemma 4 (Schwartz-Zippel Lemma). *Let $f : \mathbb{Z}_p^m \rightarrow \mathbb{Z}_p$ be a non-zero m -variate polynomial of degree d , we have*

$$\Pr[f(\mathbf{x}) = 0 : \mathbf{x} \leftarrow \mathbb{Z}_p^m] \leq \frac{d}{p}.$$

Theorem 5 (Knowledge Soundness). *The protocol MatrixMul achieves computational knowledge soundness in the generic group model.*

Proof. Let $S_1 = S_2 = S_T = \{0, 1\}^{\log p}$. Initialize the label function $\text{List}_i : \mathbb{Z}_p[*] \rightarrow \{0, 1\}^{\log p}$ and introduce formal indeterminates

$$T_X, T_Y, T_Z$$

that represent polynomial variables appearing in the CRS. Throughout we treat every group label returned by Label as a *symbolic polynomial* in these indeterminates. The oracles behave symbolically as in definition B.

Let $\mathcal{A}_{\text{MatrixMul}}$ interact with oracles Label, GroupOp, and PairingOp. Define the sets of formal polynomials over $\mathbb{G}_1$ and $\mathbb{G}_2$:

$$\begin{aligned}
L_1 = & \left\{ 1, \sum_{i \in [\ell]} K_i^{(1)} L_i(T_X), \sum_{i \in [\ell]} K_i^{(2)} L_i(T_X), \sum_{j \in [t]} K_j^{(3)} \bar{L}_j(T_Y) \right\} \cup \left\{ \sum_{i \in [0, \ell]} \alpha_j^{(1)} T_X^j + \right. \\
& \sum_{i, j \in [\ell]} \beta_{i, j}^{(1)} K_i^{(1)} T_X^{j-1}, \sum_{i \in [0, \ell]} \alpha_j^{(2)} T_X^j + \sum_{i, j \in [\ell]} \beta_{i, j}^{(2)} K_i^{(2)} T_X^{j-1}, \sum_{i \in [0, t]} \alpha_j^{(3)} T_Y^j + \\
& \left. \sum_{i, j \in [t]} \beta_{i, j}^{(3)} K_j^{(3)} T_Y^{i-1} \right\} \cup \left\{ \overbrace{\sum_{i \in [0, \ell]} \hat{\alpha}_j^{(1)} T_X^j + \sum_{i, j \in [\ell]} \hat{\beta}_{i, j}^{(1)} K_i^{(1)} T_X^{j-1}}^{Q_{XY}}, \sum_{i \in [0, \ell]} \hat{\alpha}_j^{(2)} T_X^j + \right. \\
& \sum_{i, j \in [\ell]} \hat{\beta}_{i, j}^{(2)} K_i^{(2)} T_X^{j-1}, \sum_{i \in [0, t]} \hat{\alpha}_j^{(3)} T_Y^j + \sum_{i, j \in [t]} \hat{\beta}_{i, j}^{(3)} K_j^{(3)} T_Y^{i-1} \left. \right\} \cup \\
& \left\{ \overbrace{\sum_{i \in [0, \ell]} \bar{\alpha}_j^{(1)} T_X^j + \sum_{i, j \in [\ell]} \bar{\beta}_{i, j}^{(1)} K_i^{(1)} T_X^{j-1}}^{q_{XY}}, \sum_{i \in [0, \ell]} \bar{\alpha}_j^{(2)} T_X^j + \sum_{i, j \in [\ell]} \bar{\beta}_{i, j}^{(2)} K_i^{(2)} T_X^{j-1}, \right. \\
& \left. \sum_{i \in [0, t]} \bar{\alpha}_j^{(3)} T_Y^j + \sum_{i, j \in [t]} \bar{\beta}_{i, j}^{(3)} K_j^{(3)} T_Y^{i-1} \right\}, \\
& \text{where } K_i^{(1)} = \sum_{j \in [t]} \kappa_{i, j}^{(1)} \bar{L}_j(T_Y), K_i^{(2)} = \sum_{k \in [s]} \kappa_{i, k}^{(2)} \tilde{L}_k(T_Z), K_j^{(3)} = \sum_{k \in [s]} \kappa_{j, k}^{(3)} \tilde{L}_k(T_Z).
\end{aligned}$$

and

$$L_2 = \left\{ 1, \overbrace{\sum_{i \in [\ell]} \delta_i^{(1)} L_i(T_X)}^{r(T_X)}, \overbrace{\sum_{j \in [t]} \delta_i^{(2)} \bar{L}_j(T_Y)}^{h(T_Y)} \right\} \cup \{h_Y, d_Y\} \cup \{T_X^\ell - 1, T_Y^t - 1\}.$$

Here, each label corresponds to a formal polynomial in the unknown scalars τ_X, τ_Y, τ_Z . For compactness, keep the Lagrange basis polynomials $L_i(T_X), \bar{L}_j(T_Y), \tilde{L}_k(T_Z)$ as fixed polynomials in the respective variable.

The pairing equations are linear in $L_1 \otimes L_2$:

$$\begin{aligned}
e(V, [r(\tau_X)]_2) &= e([h(\tau_Y)]_2, [1]_2) \cdot e(D_{XY}, [\tau_X^\ell - 1]_2) \cdot e(Q_{XY}, [\tau_X]_2), \\
e(U, [r(\tau_X)]_2) &= e([d(\tau_Z)]_2, [1]_2) \cdot e(\bar{D}_{XZ}, [\tau_X^\ell - 1]_2) \cdot e(\bar{Q}_{XZ}, [\tau_X]_2), \\
e(P, [h(\tau_Y)]_2) &= e([d(\tau_Z)]_2, [1]_2) \cdot e(\tilde{D}_{YZ}, [\tau_Y^t - 1]_2) \cdot e(\tilde{Q}_{YZ}, [\tau_Y]_2), \\
e(Q_{XY}, [\tau_X]_2) &= e(q_{XY}, [1]_2), e(\bar{Q}_{XZ}, [\tau_X]_2) = e(\bar{q}_{XZ}, [1]_2), \\
e(\tilde{Q}_{YZ}, [\tau_Y]_2) &= e(\tilde{q}_{YZ}, [1]_2).
\end{aligned}$$

which implies the following *symbolic* identities hold:

$$\begin{aligned} V(T_X, T_Y) \cdot r(T_X) &= h(T_Y) + D(T_X, T_Y) \cdot (T_X^\ell - 1) + Q(T_X, T_Y) \cdot T_X, \quad Q(T_X, T_Y) \cdot T_X = q(T_X, T_Y) \\ U(T_X, T_Z) \cdot r(T_X) &= d(T_Z) + \bar{D}(T_X, T_Z) \cdot (T_X^\ell - 1) + \bar{Q}(T_X, T_Z) \cdot T_X, \quad \bar{Q}(T_X, T_Z) \cdot T_X = \bar{q}(T_X, T_Z) \\ P(T_Y, T_Z) \cdot h(T_Y) &= d(T_Z) + \tilde{D}(T_Y, T_Z) \cdot (T_Y^t - 1) + \tilde{Q}(T_Y, T_Z) \cdot T_Y, \quad \tilde{Q}(T_Y, T_Z) \cdot T_Y = \tilde{q}(T_Y, T_Z). \end{aligned}$$

We use the Lemma 3 to the identities above, we have

$$\begin{aligned} &\sum_{i \in [\ell]} \delta_i^{(1)} K_i^{(1)} + D(T_X, T_Y) \cdot (T_X^\ell - 1) + Q(T_X, T_Y) \cdot T_X \\ &= h_Y + \left(\sum_{i \in [0, \ell]} \alpha_j^{(1)} T_X^j + \sum_{i, j \in [\ell]} \beta_{i, j}^{(1)} K_i^{(1)} T_X^{j-1} \right) \cdot (T_X^\ell - 1) + \\ &\quad \sum_{i \in [0, \ell]} \bar{\alpha}_j^{(1)} T_X^j + \sum_{i, j \in [\ell]} \bar{\beta}_{i, j}^{(1)} K_i^{(1)} T_X^{j-1}. \end{aligned}$$

where

$$\left(\overbrace{\sum_{i \in [0, \ell]} \hat{\alpha}_j^{(1)} T_X^j + \sum_{i, j \in [\ell]} \hat{\beta}_{i, j}^{(1)} K_i^{(1)} T_X^{j-1}}^{Q_{XY}} \right) \cdot T_X = \overbrace{\sum_{i \in [0, \ell]} \bar{\alpha}_j^{(1)} T_X^j + \sum_{i, j \in [\ell]} \bar{\beta}_{i, j}^{(1)} K_i^{(1)} T_X^{j-1}}^{q_{XY}}.$$

We observe that $\beta_{i, j}^{(1)} = 0$ for all i, j since $\beta_{i, j}^{(1)} K_i^{(1)} T_X^{j+\ell}$ does not appear anywhere in this target equation. Furthermore, $\bar{\beta}_{i, j}^{(1)} = 0$ for all i, j . Thus,

$$\delta_i^{(1)} = 0, \quad \text{for each } i \in [\ell].$$

This gives us that $r(T_X) = \sum_{i \in [\ell]} \delta_i^{(1)} L_i(T_X) \equiv 0$ is a constant function, which is a contradiction. Furthermore, $h(T_Y) \equiv 0$.

Bad Event. When instantiating the generic group with a concrete bilinear group of prime order p , it is possible that two distinct formal polynomials evaluate to the same group element under a random assignment of $\tau_X, \tau_Y, \tau_Z \in \mathbb{Z}_p$. By the Schwartz–Zippel lemma 4, for any two distinct polynomials f and g of total degree at most $D = \max(\ell - 1, t - 1, s - 1)$, we have

$$\Pr[f(\tau_X, \tau_Y, \tau_Z) = g(\tau_X, \tau_Y, \tau_Z)] \leq \frac{d}{p}.$$

There are $O(\ell t + \ell s + ts + Q)$ distinct formal polynomials that may appear in the adversary's view, each of degree at most d . By union bound, the probability of this “bad event” is

$$\Pr[\text{Bad}] = O\left(\frac{(\ell t + \ell s + ts + Q)^2 \cdot \max(\ell, t, s)}{p}\right),$$

where the first three terms come from CRS elements and prover messages, and Q bounds the number of oracle queries made by the adversary. Thus, the knowledge soundness of our proof system holds in the generic group model. $\square$